

TrayGuard

A Local Tray Training And Assessment System

Final Design Review

EC ENGR 180DW – Systems Design

University of California, Los Angeles

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1 Executive Summary

(Matthew)

Tray-content errors during reconstruction of reusable stainless steel surgical instrument trays are a recurring patient-safety risk in sterile processing departments (SPDs). Root-cause analysis identified six contributing categories—training gaps, incomplete count sheets, local tray variation, nomenclature inconsistency, production pressure, and weak feedback loops—all driven by the structural absence of professional licensure and competency infrastructure in the SPD workforce.

The project initially pursued a computer-vision tray checker. Seven staged detection experiments (800–80 lumens, matte/reflective backgrounds, separated/overlay layouts) showed that YOLO11s detection degrades under every variation that SPD workflows present, with mAP50-95 falling to 0.389 under occlusion. Real-instrument experiments confirmed that position cues dominate over shape learning, making lighting-transfer conclusions unreliable without randomized layouts. Deployment-grade recognition would require site-specific training data, per-manufacturer validation, and workflow testing beyond the project’s available resources.

The project pivoted to a training-centered platform: TrayGuard, a local tray training and assessment system using printable AprilTag instrument cards and a seven-step learning loop (intake, pre-test, study cards, quiz, practice sort, post-test, summary). A file-backed tray module defines instruments, aliases, required counts, distractors, lookalike pairs, and assessment variants. A scoring engine classifies each item into correct, missing, extra, wrong, misidentified, and wrong-count categories with confidence metadata.

A pre/post study with the seeded tray module would provide Kirkpatrick Level 2 evidence (simulated learning) with bounded claims. The primary limitation is the use of student participants and physical card proxies—workplace transfer and clinical adoption remain future validation. The recommended next step is to build, test, and report the prototype study within the current scope.

2 Introduction And Project Overview

(Matthew)

Tray content errors during surgical instrument reconstruction are recurring operational failures that can delay surgery and compromise patient safety. The workforce faces structural pressure from growing demand, high turnover, and limited training capacity. Current training blends online coursework, classroom instruction, and hands-on externships that are capacity-constrained by clinical-site availability, clinical supervisor attention, and local teaching materials.

TrayGuard originally targeted a computer-vision-based tray checker: a camera above the assembly workstation would detect each instrument as the technician placed it, compare detections against the tray list, and return a go or no-go verdict before sealing. The CV approach failed early because it required the project to validate detection accuracy, lighting tolerance, occluded-item handling, and workflow integration inside a live SPD environment, resources not available within our project scope. The project pivoted to a training-centered alternative that preserves the same error categories (missing, wrong, extra, misidentified, and miscounted items) but measures them in a simulated local tray task using AprilTag instrument cards, retrieval study, and feedback modes.

This Functional Design Review serves as the literature-grounded design handoff for a future implementation team. It defines the problem, justifies the training-centered solution, specifies requirements and system architecture, provides a study plan for novice evaluation, and documents the scope boundaries that prevent overclaiming.

3 Problem Definition And Scope

(Matthew, Andy, Qiyu)

3.1 Precise Problem Statement

Errors can occur throughout the sterile processing workflow: a technician may confuse two visually similar instruments, place the wrong size or pattern into a tray, omit or add an instrument, use the wrong count sheet, or fail to catch a documentation problem. In the OR, those mistakes appear as missing, wrong, extra, damaged, contaminated, or otherwise unavailable instruments.

Published rates vary by hospital, measurement method, and error definition, but the available evidence suggests that these are recurring operational failures, not isolated accidents. Alfred et al. analyzed 3,900 tray defects across 41,799 cases, or roughly 9 defects per 100 cases, and found that 55.0% of recorded defects occurred during assembly [1]. Zhu et al. found 398 packaging errors among 33,839 surgical instrument packages, or about 1.2% of packages. A detailed breakdown reveals nine categories: instrument in wrong specification (right type, wrong variant; 44.0%), instrument missing (19.4%), package incomplete (17.6%), instrument malfunction (6.8%), wrong packaging material (3.8%), box and cover mismatched (3.5%), wrong packaging tag (2.0%), indicator card missing (1.5%), and wrong count of instrument (1.5%) [2]. Nichol et al. observed 236 surgical instrument errors affecting 147 cases, with most errors tied to visualization work such as inspection, identification, function checking, and tray assembly. The resulting OR delays averaged 10.16 minutes per affected case, at an estimated annual cost of \$6.75 million to \$9.42 million [3]. Nichol and Saari further identified 368 patient safety notices containing 419 surgical instrument errors over 12 months, of which 83% involved failures in inspection, tracking, or identification [4].

The general problem is therefore:

Sterile processing workflows do not always reliably produce complete, correct, functional, sterile, and ready-for-use surgical instrument sets.

3.2 Practice Context

The problem centers on reusable stainless steel surgical instruments, which make up the majority of SPD-handled inventory. Unlike single-use devices or textile goods, these instruments must be visually identified, sorted, and matched to tray lists after every use, and their metallic surfaces, subtle dimensional differences, and reprocessing wear make them especially susceptible to the identification errors documented below. George et al. describe a simplified SPD-to-OR cycle [5]:

1. Point-of-use preparation: at the end of a procedure, instruments are prepared for transfer from the OR to sterile processing. This can include gross cleaning, use of sterile water, and enzymatic pretreatment to keep material from drying on instruments.
2. Transport to sterile processing: used instruments are moved from the OR or procedural area to the sterile processing unit.
3. Decontamination: instruments are cleaned and decontaminated. Manual scrubbing, ultrasonic cleaning, and automated washer systems are common parts of this stage.
4. Assembly and inspection: after decontamination, instruments move to the assembly area. Technicians inspect instruments, assemble instrument pans or sets, replace missing items, and place instruments into containers or wraps for sterilization.
5. Sterilization: packaged instruments are sterilized using methods appropriate to the device and material. High-temperature methods such as autoclave sterilization and low-temperature methods such as chemical or radiation-based sterilization are used.

6. Storage: once sterilized, instruments and trays are cooled as needed and placed in storage until they are required again.
7. Return to use: when needed for surgery, stored instruments are obtained and brought back to the OR for the next case.

3.3 Narrowed Problem

Within the full SPD-to-OR cycle, this project narrows to tray reconstruction and verification: the point where cleaned instruments are inspected, identified, matched to count sheets, and rebuilt into procedure-specific trays. Nichol and Saari built a risk model of the full surgical instrument cycle because published error studies had documented that defects occurred but had not systematically mapped where in the instrument lifecycle—from OR use through cleaning, assembly, sterilization, storage, and return—errors were most likely. Their model, which studied reusable stainless steel surgical instruments, decomposed the cycle into 104 human tasks and found that 91% of modeled error risk occurred during the sterile-processing portion, especially the decontamination and assembly-and-inspection steps. All eight highest-risk tasks involved human visualization, sorting, and local tray knowledge [6].

3.4 Stakeholders And Importance

Instrument readiness matters because surgery is time-sensitive, coordination-heavy work. When an instrument set is wrong, incomplete, unclean, damaged, or unavailable, the OR team may need to pause, search for replacement instruments, open backup trays, change the procedure flow, or delay the case [3], [7]. Dirty or damaged instruments add a direct safety concern. A 2024 hospital inspection reported pitting (surface corrosion), stains, sticky residue, rust, scratches, and other concerns in randomly selected ready-for-use trays [8]. The Pennsylvania Patient Safety Authority warns that contaminated instruments can put patients at risk of surgical site infection and can also cause lost OR time if discovered after a procedure has begun [7]. SPD backlogs have escalated beyond individual delays: in 2025, Colorado’s largest hospital paused all surgeries for a week after inspectors found pitting (surface corrosion), stains, and residue in randomly selected ready-for-use trays [9]. The stakeholders include:

- Patients, who depend on sterile, functional instruments during surgery.
- Surgeons and OR teams, who need complete and usable sets to maintain procedure flow.
- SPD technicians, who perform detailed, high-volume work under production pressure.
- SPD supervisors and educators, who manage training, quality assurance, staffing, documentation, and feedback loops.
- Hospitals, which lose time and capacity when instrument issues delay cases or require rework.

3.5 General Interest Beyond One Local Practice

Most people will undergo surgery at least once [10]. Each procedure depends on correctly assembled, sterile instrument trays, which means each procedure depends on the same manual identification and reconstruction task that the sections above document as vulnerable. The problem is not confined to one hospital’s workflow; it is embedded in every OR that relies on prepared instrument sets.

3.6 Root Cause Analysis

The fishbone head for this analysis is:

Tray content errors during reconstruction of reusable stainless steel instrument trays.

Product.

Same-family instruments (Kelly/Crile, straight/curved Mayo) differ only in subtle dimensional features with no intrinsic labels or markings [1]; [11]; [3]. For example, a tray requiring four 14 cm Kelly clamps and two 16 cm Kelly clamps can be misassembled as two 14 cm and four 16 cm clamps—the total count is correct but the instrument variant is wrong, a pattern that Zhu et al. found accounted for 44.0% of all instrument packaging errors [2]. Steam autoclaving destroys adhesives and fades inks, laser etching is unstandardized across manufacturers, and reprocessing wear further reduces visible cues [12]; [13].

The instrument supply chain evolved around surgical function, not SPD identification. Manufacturers bear labeling cost, and SPD receives the benefit — but SPD has no purchasing authority over instrument design [14]. No stakeholder has enough individual incentive to change instrument-marking practices because error costs are small incremental delays spread across the workflow, not a direct impact on surgery or surgical liability [8]. No regulatory body mandates marking because no sentinel event with a clear causal line to patient death has been attributed to unlabeled instruments [15]. Retrofitting across existing inventory is prohibitive.

Tray composition, instrument design, inventory, and unstandardized nomenclature make each tray a complex product to reconstruct reliably [1]. Tray contents are local design decisions rather than fixed universal lists. Procedure-specific, surgeon-specific, and hospital-specific decisions determine which instruments and quantities belong in a set [16]; [17]. A user can know instrument families and still select the wrong item or quantity for a given tray. Reconstruction must also include rejection of visibly unacceptable items, not only selection of named items. Otherwise damaged or suspect instruments can remain in a tray despite correct identity and count [18]; [7].

Knowledge.

Three sub-causes converge. First, knowing the general name is not sufficient. Technicians must see the difference between nearly identical instruments [19]; [3]. Visual discrimination is a perceptual skill that requires deliberate practice with feedback [20]; [21], but the SPD workplace provides no structured opportunity for this practice. Preceptors train on the fly, production pressure prevents deliberate study, and no mechanism identifies which visual discriminations an individual has not yet mastered [3]; [1]. Certification failure rates of 28–50% confirm that the technical demands are substantive, yet the role remains classified and compensated as entry-level support labor [22].

Second, general instrument knowledge and local tray rules are separate domains. Manufacturers produce physically different tools under the same generic name [23]. Tray composition is driven by surgeon preference, hospital case mix, and historical accretion rather than a universal standard [16]; [17]. Local aliases accumulate organically on count sheets with no central nomenclature authority [1]. Count sheets encode tray-specific rules that exist only in that hospital’s workflow [23].

Third, certification proves a starting point rather than local readiness. Neither HSPA nor CBSPD mandates recurring competency checks tied to local tray knowledge [22]; [24]; [25]; [26]; [27]. Training inconsistency persists because no single role owns the standard curriculum for each tray. Differences across clinical supervisors, shifts, and training tenures produce conflicting mental models of the same tray [1]; [28]; [29].

Process.

Building a tray forces picking, counting, checking, and arranging into one step with no double-check system. Alfred et al. found that 55.0 percent of tray defects occur during assembly [1]. Nichol et al. found that 88.6 percent of observed errors involve visualization tasks such as inspection, identification, function checking, and sorting [3]. Sequential verification would double labor per tray, and no accreditation mandate from the Joint Commission, CMS, or HSPA requires a two-person check for tray assembly [1]; [30].

Productivity metrics structurally incentivize single-pass assembly. The AAMI benchmarking framework uses worked-hours-per-unit (WHPU), reinforcing throughput as the primary performance signal [31]. Instrument damage such as corrosion, stains, burrs, and worn jaw serrations can be missed when inspection relies on rapid visual checks under poor lighting [11]. Workers use only their eyes and hands, without magnification, side-by-side comparison guides, or decision-support tools at the workstation [3]. No worker can remember every item in every tray when a single hospital may manage hundreds of tray configurations, each with 30 to more than 100 instruments. Production pressure and cognitive load cause workers to break from the count sheet and rely on memory, especially for familiar trays [23]. No formal measurement of count-sheet compliance exists, so the gap between assumed behavior and actual behavior is invisible to management. Tray errors have not crossed the sentinel-event threshold needed to trigger regulation [15].

Information.

Count sheets become incomplete or outdated because tray contents change continuously. Instruments are added, removed, substituted, or replaced as surgeon preferences shift, manufacturers discontinue items, or vendors supply new loaners. No dedicated role or budget exists for count-sheet maintenance. Updating a count sheet is a manual, discretionary task that competes with production work. Without a formal change-management workflow, sheets accumulate inaccuracies across multiple revision cycles [23]. Missing or outdated identification information, IFUs, and tray listings at the workstation force workers to rely on memory [32].

Loaner trays introduce a separate information failure. They arrive with manufacturer-stocked contents and vendor documentation that uses the vendor’s naming conventions and part numbers, not the hospital’s local aliases or count-sheet format. SPD receives the tray with hours or minutes to process it before surgery, and the worker must reconcile the vendor’s nomenclature against the local count sheet with no lead time to prepare or cross-reference [33].

Environment.

Multiple environmental conditions reduce the attention and time available for tray reconstruction. Pressure and interruptions break focus for detail-heavy visual work [1]; [34]. The open-plan workstation layout makes it difficult to shield assembly workers from interruptions. Phones ring with OR requests, surgeons call for add-on instruments, and runners deliver and collect trays [34]. Uncomfortable workstations, awkward reaches, and poor setup make a job that already needs close attention harder [35]. Ergonomic improvements compete for facilities budget with clinical areas that have more organizational visibility. SPD staff have less organizational voice than clinical departments [36].

The wage level is a structural cause that amplifies every other category. SPD is a cost center within the hospital. It generates no direct revenue, so staffing budgets are structurally constrained relative to OR capacity, which is a revenue center [37]; [38]. Hospitals routinely add surgery suites without proportionally increasing SPD headcount [9]. Average SPD pay is \$69,217, with a 14 percent drop in respondents receiving raises. SPD workers describe wages as “less than your local Starbucks wage” [39]; [40]. Fewer staff means less time for training and less thorough checking [1]; [25]; [41]; [42].

Three structural mechanisms—historical classification as materials management, absence of professional licensure, and low union representation—produce this wage level. Together they create a self-reinforcing cycle: structural underinvestment produces low wages, low wages produce chronic vacancies, vacancies increase production pressure, and production pressure preempts the training and process improvements that could reduce errors [43]; [44]; [45].

Policy & Feedback.

SPD governance is decentralized across shifts. No single role is responsible for maintaining, auditing, or enforcing count-sheet accuracy or tray-assembly procedure across all shifts [23]; [46]. Creating a single

authoritative standard for each tray requires a designated owner, a maintenance process, and enforcement across shifts, but none of these are budgeted.

Certification bodies (HSPA, CBSPD) recommend but do not mandate recurring competency checks tied to local tray knowledge. When competency programs exist, they are locally administered with no external audit [26]; [24]. The absence of a mandate means that even published, evidence-backed training programs — such as the Ofstead et al. (2023) borescope inspection pilot — remain discretionary investments that compete for budget with production headcount, rather than required practice.

Error reporting is manual, slow, and deprioritized. Errors discovered during surgery are documented, if at all, in incident reports that may take weeks to reach SPD management. By that time the specific tray, technician shift, and training gap are hard to identify, so the error teaches no preventive lesson [3]; [2]. The hospital incident reporting system was designed for clinical errors such as medication errors, falls, and wrong-site surgery. Tray-content errors do not fit neatly into the reporting categories, so they are underreported. Without closed-loop tracking that ties an error back to a root cause, downstream detection does not drive upstream improvement [3]; [2].

Structured process-improvement approaches can substantially reduce these defects. Using Six Sigma methodology, Natarus et al. improved first-pass yield from 81.0% to 97.4% and reduced the tray defect rate from 2.2% to less than 0.1% in a hospital sterile-processing department, demonstrating that targeted workflow and training interventions can close the gap described above [47].

Consolidated Root Cause Chain.

Sterile processing evolved from materials management, producing a logistics-oriented workforce without the professional infrastructure of clinical departments. Compensation and professional infrastructure reflect that origin: below current technical demands. Hospital associations resisted mandatory certification because it would create upward wage pressure. Low union representation leaves no institutional mechanism to counter cost-center budget logic. Temporary contract SPD technicians earn 40 to 80 percent more than permanent staff [39]; [44].

Compounding these workforce conditions, the hospital financial structure separates OR revenue from SPD cost: every SPD investment is visible as expense while error reduction has no standard metric. Hospitals add surgery suites without increasing SPD headcount, making underinvestment rational at each hospital because error costs are diffuse workflow friction.

A third structural driver compounds the first two: no information architecture connects OR outcomes to SPD actions, so upstream errors have no downstream feedback.

The pattern is visible in SPD training evidence. Ofstead et al. (2023) demonstrated that structured borescope inspection training for sterile-processing professionals produces large, durable knowledge gains: mean test scores rose from 41% to 84% after the workshop and remained at 90% after two months, and trainees identified actionable defects on patient-ready endoscopes at their own facilities [24]. Despite publication in a peer-reviewed infection-control journal (the *American Journal of Infection Control*) and endorsement by the HSPA Foundation, no mechanism exists to compel or incentivize adoption of this training model. The intervention demonstrates educational efficacy (Kirkpatrick Level 2), but the adoption decision depends on regulatory mandate, liability risk, or visible cost avoidance (Level 4), conditions that the structural drivers above keep absent. This pattern of strong learning evidence with no adoption pathway is the same dynamic that any future training tool, including TrayGuard, must anticipate.

A cross-profession comparison confirms that these structural drivers, not the quality of available evidence, explain the absence of reform. Respiratory therapy followed the canonical professionalization trajectory: the American Association for Respiratory Care (AARC) established a voluntary certification in 1969, unified credentialing under the National Board for Respiratory Care in 1974, and then lobbied

state legislatures one by one; California passed the first licensure law in 1982, and by 2004 all 48 contiguous states had enacted licensure for respiratory therapists, enforced by state boards with disciplinary authority [48]. Surgical technology is mid-trajectory: the Association of Surgical Technologists (AST) has driven over a dozen states to require graduation from a CAAHEP-accredited program and Certified Surgical Technologist (CST) credentialing, though many states still lack regulation [49].

Sterile processing followed none of this path, and the reason is not that its technical demands are lower. A comparison of enabling conditions makes the gap explicit:

Table 1: Enabling conditions for professionalization: respiratory therapy, surgical technology, and sterile processing

Enabling condition	Respiratory therapy	Surgical technology	Sterile processing
Visible patient harm creating political will	Respiratory-failure patient deaths drove legislative urgency	Surgical-site infection outbreaks provided the harm narrative	Tray-content errors have not crossed the sentinel-event threshold [15]
Professional association with lobbying infrastructure	AARC maintains full-time government-affairs staff and drafts model legislation [48]	AST runs a sustained state-by-state legislative campaign [49]	HSPA and CBSPD are voluntary certification bodies, not advocacy organizations
Standardized education pipeline	CoARC-accredited programs are the required entry path	CAAHEP-accredited programs required by law in regulated states	No mandated program accreditation; 400 hands-on hours with no curriculum standard
Direct patient-care lineage	Evolved from ICU respiratory support (1950s–60s)	Evolved from intraoperative assistance in the OR	Evolved from central supply and materials management [14]

Each condition maps to a structural driver identified above. The absence of visible harm corresponds to the broken feedback loop that keeps error costs invisible. The absence of lobbying infrastructure and education standardization corresponds to the weak professionalization that followed from SPD’s materials-management origin. The absence of a patient-care lineage corresponds to the cost-center classification that suppresses wages and organizational voice. Together these explain why a role with comparable technical demands remains a semi-profession while adjacent roles achieved state-enforced regulation [50].

No information architecture connects downstream OR outcomes to upstream SPD actions. The feedback loop from error to correction is broken, so the same failures repeat without triggering systemic change. Error reporting is designed for clinical events, not tray-content errors.

The system remains stable because no stakeholder has both the incentive and the authority to change it. Manufacturers control instrument design but bear no cost from SPD identification errors. Hospital administrators control SPD budgets but see wage increases as expense and error costs as invisible friction. SPD workers perform the work but lack the union representation, professional licensure, or organizational voice to demand structural change. Regulatory bodies have not acted because tray errors have not crossed the sentinel-event threshold.

4 Literature And Related Work

(Matthew, Andy, Qiyu)

4.1 The Solution Landscape

The solution space for reducing tray-content errors during reconstruction can be organized into seven categories by intervention point. Each category targets a different subset of the root causes, and each

faces a different feasibility constraint for this project.

The seven identified solution categories are:

1. **Automated content verification:** post-assembly inspection using CV, RFID, or barcode scanning to verify tray contents.
2. **Physical error-proofing:** modifying the tray or instrument interface (custom foam, color coding) to prevent selection errors.
3. **Workflow / process design:** restructuring task handoffs and verification steps through dual checks or gating sign-offs.
4. **Decision support at point of assembly:** providing standardized count sheets, photos, and alias lookup at the workstation.
5. **Task simplification:** rationalizing tray contents and standardizing instrument sets across procedures.
6. **Workforce / accountability systems:** competency reassessment, closed-loop error tracking, and structured mentorship.
7. **Training and simulated practice:** building local tray skills through retrieval practice, simulation, and feedback before supervised work.

Table 2: Seven solution categories compared by constraint

Category	Intervention point	Example solutions	Root causes addressed	Key constraint for this project
Automated content verification	Post-assembly inspection	CV tray checker, RFID tray scan, barcode line scan, weight check	Product: wear hides differences and Process: no backup check exist	Requires real instruments, lighting, workflow integration; cannot validate without SPD deployment
Physical error-proofing	The tray or instrument interface	Custom foam cutouts, color-coded handles, segregated tray zones	Knowledge: workers must memorize tiny visual details and Process: assembly lacks a separation of pick from check	Expensive custom fabrication per tray; difficult to validate and maintain across tray variants
Workflow / process design	Task structure and handoffs	Dual verification, gating sign-offs, pre-sorted pick lists	Process: no structured backup check during assembly and Policy: nobody formally owns the standard	Primarily organizational change; no technical prototype to build or measure in a controlled study
Decision support at point of assembly	Information available during re-construction	Standardized count sheets with photos, alias lookup, reference kiosk	Information: outdated count sheets force guessing and Knowledge: general instrument names do not convey local tray rules	Improves the reference but does not exercise or measure the technician's own ability
Task simplification	The tray definition itself	Tray rationalization, instrument consolidation, procedure-level standardization	Knowledge: differentiating lookalikes requires local familiarity and Environment: pressure and interruptions break focus	Requires hospital utilization data and multi-stakeholder buy-in; no buildable prototype
Workforce / accountability systems	Competency management and error feedback	Periodic competency reassessment, closed-loop error tracking, structured mentorship	Policy: standards are only tested at hiring and Feedback: slow error reporting lets mistakes repeat	Primarily organizational policy change; no measurable prototype within our project scope
Training and simulated practice	Pre-work and between-work skill building	VR simulation, digital flashcards, tangible tray simulation, structured OJT	Knowledge: lookalike discrimination requires repeated practice and Local knowledge: tray-specific names and rules are not taught in general curricula	Requires learning-science justification and a novice study, both accessible without SPD deployment

Each category reduces a real risk. Categories 1 through 6, however, all depend on resources that this

project cannot access within our project timeline: SPD deployment sites, real instrument inventories, hospital utilization data, institutional policy authority, or multi-stakeholder buy-in. Category 7 – training and simulated practice – is the only category whose primary evidence can be collected with accessible participants (novices), low-cost materials (printed cards), and a bounded pre/post study design on a simulated local tray task. Section 4 examines the literature supporting each relevant area in depth.

4.2 SPD Training and Competency Gap

SPD training in practice follows a blended pattern that combines coursework, hands-on practice, and clinical supervisor-led onboarding:

Table 3: SPD training patterns and TrayGuard implications

Training pattern	Evidence	Implication for TrayGuard
Certification exam plus verified hands-on hours	HSPA CRCST requires 400 SPD hands-on hours and a computer-based exam; CBSPD CSPDT requires 12 months of full-time SPD employment (or a training course with passing grade of 70 or higher, or 6 months allied health plus 6 months SPD experience) [22]; [51].	The tool should measure learning and support practice before and during supervised competency development.
Self-paced or classroom instruction	Towson's ed2go-backed course is self-paced online, 190 hours, covering instrument identification and tray assembly. Altamont's program is 135 hours over 8-12 weeks with live online classes, in-person labs, and an optional 240-400 hour externship [52]; [53].	The platform can complement formal instruction with reusable local modules and repeated practice.
Clinical supervisor-led onboarding	HPN describes hospital examples including 90-day on-the-job training programs, 19-week programs with textbook chapters, quizzes, and progress tests, and structured 1:1 clinical supervisor onboarding with presentations, online modules, written tests, direct observation, and one-on-one training [27]. HSPA training guidance frames core program elements as structured orientation, competency documentation, and ongoing education; training that stops after orientation perpetuates inconsistent habits [54].	The system should reduce low-level explanation burden and give clinical supervisors evidence of where a learner still struggles.
Simulation and skills assessment	AORN describes Penn Medicine's earn-to-learn SPD program combining didactic education, hands-on training, simulation, mentorship, content testing, and skills-based assessment [25].	Simulated tray sorting and pre/post tasks are aligned with existing training practice.

The common thread across patterns is that hands-on practice and clinical supervisor attention are the scarce resources. Several program examples show the gap between classroom preparation and clinical capacity. Penn Foster's online sterile processing program prepares learners for the CRCST exam but does not require the exam or the 400 hands-on hours [55]. MedCerts states that full CRCST certification requires 400 hours of hands-on experience following provisional certification [56]. AIMS Education

requires a 400-hour clinical internship, but its schedules depend on clinical-site availability [57]. Central Sterilization Solutions states its classroom course has no hands-on training in the classroom and that hands-on training is conducted during externship [58]. These examples illustrate the training-access gap: repeatable, feedback-rich practice with local instruments and tray rules is hard to scale before and during real SPD experience because it depends on site access, clinical scheduling, clinical supervisor attention, and available teaching materials.

Structured training programs for sterile-processing staff have shown positive results. Hu et al. developed a CSSD training program using action research; after two rounds of training, CSSD nurses showed significant improvement in professional knowledge [28]. Huang et al. similarly concluded that an action-research-based training program improved CSSD nurses’ professional knowledge and skill, increased their enthusiasm for learning, and laid a foundation for talent development [34]. These findings support TrayGuard’s role as a repeatable training supplement: local SPD practice varies and depends on supervised experience with site-specific trays, sheets, and policies.

Broader allied-health placement research supports treating clinical placements as capacity-constrained environments. A Queensland study found that placement-offer growth did not keep pace with student-program growth, and barriers centered on resourcing and increasing demand [59]. A 2025 Greater Sacramento report identified shortage of clinical placement opportunities as a barrier to workforce pipeline growth, and highlighted student preparedness as part of successful placement partnerships [60]. TrayGuard gives learners repeated practice before, during, or between supervised work, and gives educators evidence about weak instruments, weak tray rules, high-confidence errors, and uncertainty. This addresses the practical constraint: supervised training time is a scarce resource [25]; [42]; [41].

4.3 Local Tray and Count-Sheet Evidence

The RCA (Section 3.6) identified incomplete count sheets and local tray variation as root causes. Count sheets define tray names, sections, contents, quantities, reference numbers, placement, and sign-off fields (Nadeau, 2024; Stryker, 2023). Tray contents are design choices shaped by procedure, surgeon, and hospital policy (dos Santos et al., 2021; Ahmadi et al., 2023), so general instrument knowledge does not transfer across sites. Loaner trays add further complexity through scheduling, delivery, IFU, and count-sheet dependencies (STERIS, 2021). TrayGuard stores local names, aliases, approved photos, required counts, lookalike pairs, distractors, assessment variants, and module version. Modules must be instructor-verified before results are described as content-valid.

FGVC-derived modules serve a narrower prototype role. Atabuzzaman et al. collected ultra-fine-grained surgical instrument images from major, minor, and eye tray contexts [61]. TrayGuard’s current FGVC prototype uses the major-tray image archive and class-name mapping to create image cards and same-family discrimination exercises. The FGVC source provides observed instrument classes and representative images; public count sheets provide the field structure for tray names, sections, quantities, locations, check boxes, and sign-off fields [62]; [63]. The prototype’s FGVC quantities are simulated as one card per selected class, and exact local tray membership remains an instructor-review variable.

The scale of local variation is substantial. A multi-specialty tray optimization project at a Danish university hospital considered 1,340 different instrument trays across all surgical specialties; introducing modular trays reduced instruments in some existing trays while others required structural changes to the tray set itself [64]. In a major academic hospital in the Netherlands, a single general-surgery tray contained 94 reusable instruments used across 162 procedures in multiple disciplines, yet only 19 instruments were used per case on average and 10 were never used [65]. Tray optimization—removing rarely used instruments—reduced that tray’s size and weight by 54% [65]. These figures illustrate why general instrument familiarity is insufficient: a learner must master not only instrument shapes and names but also the specific tray configuration, quantities, and local naming conventions of their facility.

4.4 Learning Science and Simulation Evidence

Health professions education provides a general precedent for simulation: a broad review of technology-enhanced simulation found improved knowledge, skills, and behaviors compared with no intervention, and simulation-based medical education with deliberate practice has also been shown to outperform traditional clinical education in meta-analysis [66]; [67]. Earlier medical simulation guidance identifies feedback, repetitive practice, curriculum integration, and measurable outcomes as important features of effective simulation [68]. For sterile processing specifically, Ofstead et al. provide the closest direct precedent. Their borescope/endoscope visual-inspection training pilot used a pre-test, structured teaching, hands-on practice, image-based testing, confidence/satisfaction measures, workplace homework, and a delayed booster with certified sterile-processing professionals. Mean test scores improved from 41% at pre-test to 84% after the workshop, and remained at 90% after two months, with surveyed trainees reporting greater confidence after the training [24]. The electronic-flashcard evidence is especially relevant to the proposed study and card modes. Barrison et al.’s 2025 scoping review found that electronic flashcards are widely used in health-professions education, that the research base grew rapidly from 2019 to 2024, and that many studies examine utilization and associated learning outcomes. The review also cautions that development and delivery methods are less systematically studied [69]. For TrayGuard, that means flashcards are a credible medical-learning mechanism, but they should be designed as one part of a larger tray simulation: short prompts, single target concepts, local photos or card representations, immediate answer feedback, spacing or weak-item repetition, and a downstream tray-sort task that tests application rather than isolated recall. The proposed AprilTag-card tray is a reasonable physical proxy because it preserves the target cognitive actions while replacing regulated instruments with safe, cheap, repeatable tokens. The user still has to read a local tray list, retrieve which items belong, distinguish distractors and lookalikes, apply counts, place items into a tray area, and respond to feedback. The marker makes the simulated choice observable to the app. NeuroVase is a close emerging design precedent: it uses tangible cue cards with a tablet-based mobile AR system, structured medical curriculum, pre/post assessment, usability measures, and a controlled user study for neurovascular anatomy and stroke education [70]. QR-code education literature similarly supports low-cost printable codes as an access layer for healthcare learning, simulation, and training support [71]. The same design pattern appears in workplace training research. Meta-analytic evidence shows that organizational training improves learning, behavior, and operational results when it includes practice and feedback [72], and Salas et al. argue that effective training should be treated as a system: analyze the work, design instruction around target competencies, provide practice and feedback, and evaluate transfer [73]. TrayGuard follows that model by targeting a specific work task (tray reconstruction), designing instruction around instrument identity and count-sheet rules, providing repeated practice with error-specific feedback, and measuring pre/post transfer to a comparable tray variant. Serious tabletop and board-game precedents also support the choice to turn a healthcare protocol into a bounded physical learning game. Ward et al. designed and evaluated the PlayDecide patient-safety board game in two acute teaching hospitals; the intervention used cards and facilitated discussion to teach junior doctors about safety concerns and reporting, and the authors concluded that it was valuable for patient-safety education and deep discussion [74]. Their study demonstrates that abstracted physical artifacts can stand in for clinical objects when the study measures learning, discussion, decision quality, and protocol use. Marker cards are the first prototype for measuring novice improvement on simulated local tray sorting. A follow-on SPD pilot would replace or supplement cards with local photos, real teaching instruments, educator-reviewed count sheets, and supervised workplace validation.

4.5 Market and Comparable Tools

Comparable systems cluster into four groups: tray-management products, instrument tracking systems, surgical simulation tools, and emerging CV or robotic research systems. None of them eliminates the need for TrayGuard’s bounded training claim.

Table 4: Comparable tools and gap relative to TrayGuard

Tool or approach	Target user	Relevant features	Evidence strength	Gap relative to TrayGuard
Tray Pacer	SPD teams and managers	Count sheets, tray assembly support, photos, productivity and proficiency metrics.	Commercial product evidence.	Supports operations, but public materials do not establish the specific pre/post local tray learning study proposed here [75].
LayerJot SID	SPD and surgical instrument teams	Instrument documentation, photos, count sheets, and tray information.	Commercial product evidence.	Strong local-information precedent, but not a demonstrated tag-card novice learning loop [76].
CensiTrac and similar tracking systems	Hospitals, SPD, OR inventory teams	Instrument and tray tracking, traceability, inventory visibility.	Commercial and operational precedent.	Addresses tracking and lifecycle visibility more than low-cost novice simulation and weak-item training [77].
Touch Surgery and surgical simulators	Medical trainees and clinicians	Repeated simulation attempts, performance logging, skill practice.	Published evaluation and commercial adoption.	Supports simulation logic, but focuses on surgical procedure skills rather than SPD tray reconstruction [78]; [79].
NeuroVase-style tangible cards	Health-professions learners	Physical cue cards, mobile AR, structured curriculum, pre/post testing, usability measures.	Emerging preprint evidence.	Strong design precedent for tangible medical learning, but not sterile-processing content [70].
Surgical-instrument CV research	Researchers and future technical teams	Detection, counting, fine-grained classification, instrument-stand recognition.	Published research, usually bounded datasets.	Supports future camera assistance, but not deployment-ready SPD workflow claims [80]; [61]; [81].
Autonomous tray assembly research	Robotics researchers and future automation teams	Manipulation, sorting, and assembly concepts.	Early research.	Much higher hardware, sterility, safety, and integration burden than the project’s training loop [82].

TrayGuard’s contribution is a low-cost local training loop that turns a tray module into retrieval-first study, applied tray sorting, immediate feedback, no-hints assessment, and exportable pre/post evidence.

Medical training software already uses the same design pattern of simulation, guided practice, feedback, scoring, and progress tracking. Touch Surgery provides mobile surgical procedure simulations with repeated attempts and scoring, and a validation study found that users improved across repeated attempts while higher-experience cohorts outperformed lower-experience cohorts [78]. Osso VR demonstrated in a randomized trial that VR-trained novices completed a downstream orthopedic procedure more often, made fewer incorrect steps, and finished faster than guide-only learners [83]. Body Interact uses screen-based virtual patient scenarios with immediate feedback, auto-grading, and educator analytics [84]. Surgical Science and Simbionix provide commercial medical simulators across specialties, positioning simulation-based assessment as a credible healthcare training category [85]. These precedents support treating TrayGuard as a health-professions simulation tool that applies established learning mechanisms to a new domain: local SPD tray reconstruction.

4.6 Literature Gaps and System Response

The literature supports each design choice individually, but no existing work closes the specific gap that TrayGuard targets: a low-cost, repeatable, locally configurable training loop for SPD tray reconstruction that measures pre/post simulated improvement.

Table 5: Literature gaps and TrayGuard’s system response

Gap in the literature	System response	Stakeholder meaning
CV papers show instrument-detection feasibility while leaving manufacturer, site, tray, and workflow variation thinly tested [80]; [61]; [81].	Keep CV optional and locally verifiable. Use instructor-approved content as the source of truth for the training evidence.	The tool does not depend on deployment-grade CV. Camera support can be added later after site-specific validation.
SPD error papers document real defect rates while leaving the training-intervention mechanism unresolved [1]; [3].	Run a pre/post simulated learning study with error-category breakdown.	Students can test novice learnability; SPD trainees remain the future validation group for workplace transfer.
Training literature supports retrieval practice, simulation, spacing, and feedback while leaving the SPD tray-module design open [86]; [66]; [24].	Build the module around local retrieval cards, quizzes, applied sorting, weak-item repetition, and error-specific feedback.	Trainees get repeatable practice; instructors get evidence about weak items.
Adoption literature provides broad healthcare-AI guidance without SPD training-assistant specifics [87]; [88].	Report workflow friction, role boundaries, privacy assumptions, and pilot requirements alongside the prototype.	Administrators get a clearer approval path and an explicit future-evidence list.

5 First Attempt: A Computer-Vision Tray Checker (Matthew, Qiyu)

5.1 CV Literature Context

Computer vision was selected as the initial approach because published work showed that automated surgical-instrument detection and counting could work in experimental settings. Deol et al. trained a deep-learning detector on 1,004 images of 11 instrument categories and reported 98.5% precision and 99.9% recall on static images with real-time inference at 40 FPS [80]. Atabuzzaman et al. built a multi-view CNN-transformer system for 95 instrument subtypes across three surgical trays and demonstrated improved CSSD workflow efficiency in a deployed user study [61]. These papers supported CV as a candidate for tray verification.

5.2 Requirements and System Design

The original product concept used a YOLO object detector with a two-threshold confidence band to check whether a technician’s assembled tray matched the required instrument list. Detections above a high threshold were accepted as present, those between thresholds triggered a human review state, and the absence of any candidate prompted a rescan recommendation. The technician UI showed one primary action per tray state together with a visual summary of complete, missing, extra, and uncertain items, and required explicit human sign-off before any decision was final. The capture hardware was a fixed overhead camera mounted on a portable weighted board that preserved repeatable camera geometry across work surfaces. Detailed decision tables, UI requirements, and the complete workflow are documented in the following tables.

Table 6: Original planned tray-status logic using two-threshold confidence band

User-Facing State	Decision Logic
Present	Required class has at least required count of detections above high threshold T_{present} , no active similar-class or image-quality risk flag, and duplicate boxes reconciled.
Needs Review	Detections between T_{review} and T_{present} , similar class appears, count unstable, unknown or extra object plausible, or scan quality degraded.
Missing	No candidate above low threshold T_{review} , and scan quality acceptable enough that absence is meaningful.
Rescan Recommended	System lacks evidence but glare, occlusion, blur, poor framing, or lighting makes a missing label unsafe.

Table 7: Original planned technician UI requirements

Requirement	Design Direction
One primary action per state	Show the next meaningful action: start scan, rescan, review issue, confirm tray, or stop.
Clear tray-status hierarchy	Summarize Complete, Missing, Extra, and Needs Review above detailed detections.
Fast correction and rescan	Provide direct correction, dismiss, and rescan paths instead of forcing a workflow restart.
Specific review reasons	Label review prompts with the cause: low confidence, similar-class risk, glare, overlap, unknown object, or count mismatch.
Visual-first feedback with optional sound	Persistent visible status as the source of truth; short optional audio for major state changes.
Alert restraint	Escalate only conditions that change the tray decision or require user action.
Human sign-off	Require explicit final confirmation. The system supports the decision, not silently makes it.
Manual fallback	Make it clear how to continue if camera, model, or scan quality is unavailable.

5.3 CV Experiments and Results

The CV experiments tested how detection degrades when training and test conditions differ along three axes: illumination, surface reflectivity, and layout occlusion. All staged experiments used YOLO11s on a controlled self-collected dataset with four instrument-like classes (spoon variants) arranged in a fixed 11-step brightness ladder (800 to 80 estimated lumens) across two background conditions (matte, reflective) and two layout conditions (separated, overlapping). The design isolated each variable independently rather than mixing effects.

Lavado baseline. A YOLO11s model was also trained on the Lavado surgical-instrument dataset: 2,106 training images across four real instrument classes (scalpel, straight dissection clamp, straight Mayo scissors, curved Mayo scissors). After 100 epochs the model achieved precision of 0.819, recall of 0.965, mAP50 of 0.990, and mAP50-95 of 0.944 on the held-out test set (302 images). This established that the training pipeline and architecture perform competitively on a published surgical-instrument

dataset before the staged proxy experiments.

Brightness degradation. Two experiments measured monotonic mAP50-95 decay as the train-test illumination gap widened. `brightest_train_darker_test` trained on 800-lumen separated images and tested across all dimmer levels, producing mAP50-95 of 0.844. `darkest_train_brighter_test` trained on 80-lumen separated images and tested across all brighter levels, producing mAP50-95 of 0.781. The decay was asymmetric: the brightest-trained model retained higher recall under moderate drops, while the darkest-trained model recovered more quickly as light increased. Per-brightness metrics confirmed the trend was consistent across both matte and reflective backgrounds.

Background transfer. Two experiments tested whether the model learned object shape or overfit to the scene background. `matte_train_reflective_test` trained on matte-background images and tested on reflective-background images across all brightness levels, producing mAP50-95 of 0.759. `reflective_train_matte_test` trained on reflective-background images and tested on matte-background images, producing mAP50-95 of 0.830. The reflective-to-matte direction was stronger, suggesting that reflective training data provides broader feature invariance.

Occlusion stress. The separated-to-overlay layout shift produced the largest degradation. `separated_train_overlay_test` trained on non-overlapping layouts and tested on overlapping layouts, producing mAP50-95 of 0.389 compared with 0.995 on the matched-condition baseline. This gap was consistent across all brightness ranks and both backgrounds, showing that training on neatly arranged instruments does not transfer to cluttered arrangements. The overlay condition was the only experiment where recall fell below 0.70, making it the clearest product boundary: the system should request a rescan whenever tools overlap.

Table 8: Aggregate test-set metrics for the seven staged brightness experiments

Stage	Precision	Recall	mAP50	mAP50-95
<code>brightest_train_darker_test</code>	0.840	0.969	0.991	0.844
<code>darkest_train_brighter_test</code>	0.819	0.969	0.972	0.781
<code>bright_train_dim_test</code>	0.996	1.000	0.995	0.984
<code>dim_train_bright_test</code>	0.996	1.000	0.995	0.979
<code>matte_train_reflective_test</code>	0.931	0.900	0.929	0.759
<code>reflective_train_matte_test</code>	0.834	0.900	0.968	0.830
<code>separated_train_overlay_test</code> (overlay)	0.709	0.660	0.627	0.389
<code>separated_train_overlay_test</code> (baseline)	0.998	1.000	0.995	0.995

Lighting diversity (spoon proxy). A separate experiment confirmed that adding lighting variety during training improves held-out-condition detection. Three YOLO11s models were trained on the same Lavado instrument classes with increasing lighting diversity (reference lighting only, reference plus 45-degree flashlight, reference plus 45- and 90-degree flashlight). Each additional lighting condition improved precision and recall on held-out test conditions, with the most diverse training set producing the highest and most stable metrics across all held-out lighting angles.

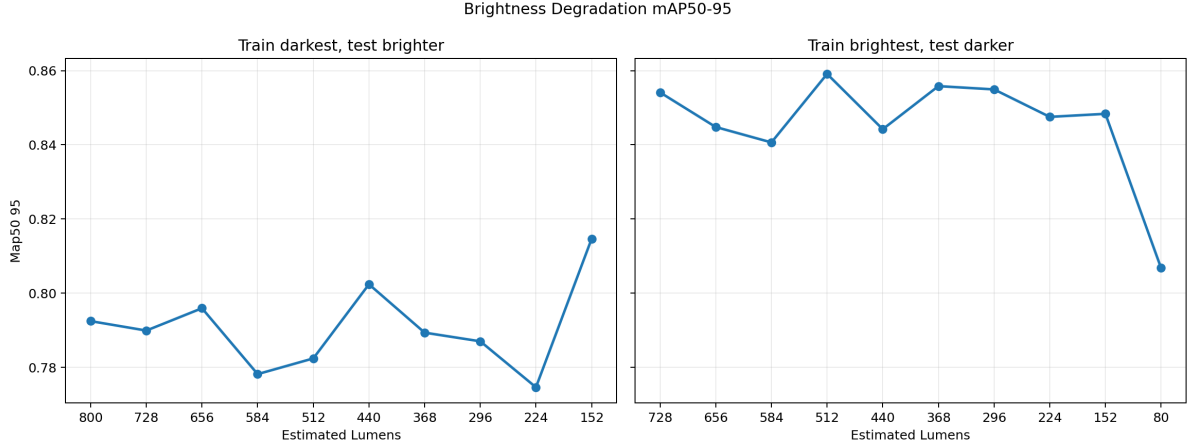


Figure 1: Brightness degradation mAP50-95 across the estimated-lumens range. Left panel: model trained on darkest (80 lumens), tested on brighter levels. Right panel: model trained on brightest (800 lumens), tested on dimmer levels.

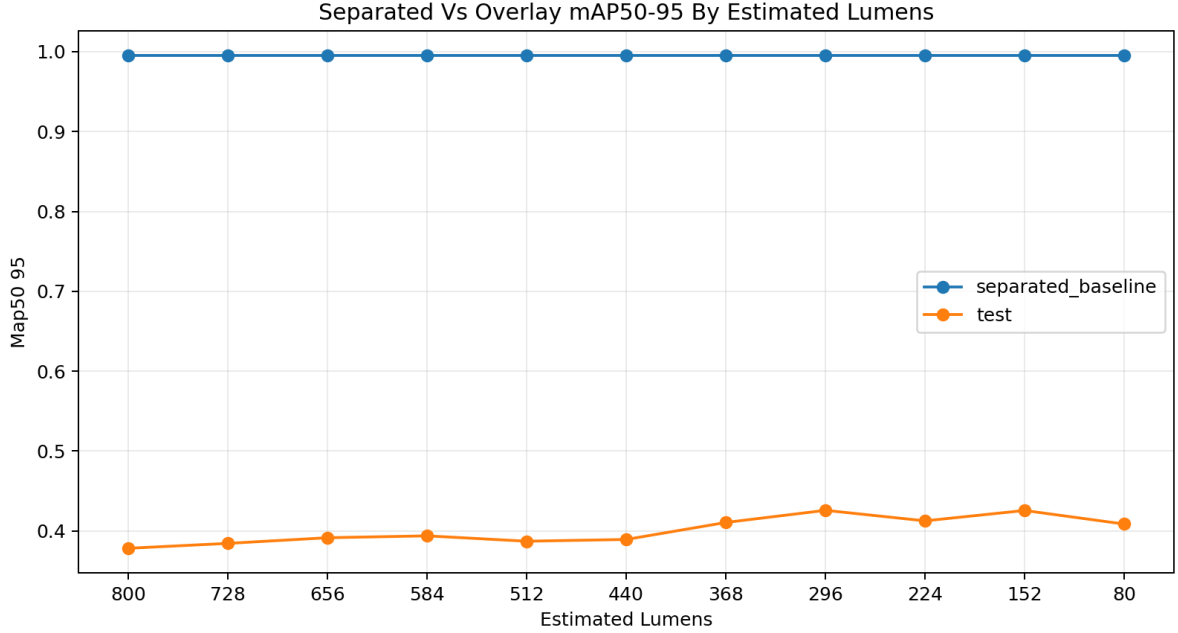


Figure 2: Separated versus overlay layout: mAP50-95 across the estimated-lumens range. The overlay condition produces the largest degradation (mAP50-95 = 0.389).

Real-instrument experiments. The brightness experiments above used a spoon proxy to measure degradation under controlled variables. A further set of experiments tested detection on real surgical instruments—six classes (scissor1–4, forcep, scalpel) from a surgical training kit photographed on matte and reflective backgrounds across four sessions. Three staged experiments were trained for 100 epochs each using YOLO11s and validated on held-out sessions:

Table 9: Aggregate test-set metrics for the three real-instrument experiments

Stage	mAP50	mAP50-95	P	R
<code>matte_train_reflective_test</code>	0.336	0.233	0.425	0.236
<code>reflective_train_matte_test</code>	0.160	0.119	0.123	0.160
<code>shape_similarity_real_proxy</code>	0.300	0.253	0.239	0.347

These scores are substantially lower than the spoon-proxy results. Per-class analysis reveals a systematic pattern: each image contains exactly six instruments in fixed positions per session, so a model can learn which classes appear at which image locations rather than learning visual features. Different classes survive under different splits—scissor1 reaches 0.862 AP under the shape-similarity split but falls to 0.000 under lighting transfer splits, while forcep and scissor2 follow the opposite pattern. Only scalpel, the most visually distinct instrument (single blade vs. multi-loop scissors), transfers across conditions (0.717 AP in reflective-to-matte). The lighting transfer experiments are therefore confounded by position-cue leakage: the test metric reflects both lighting difficulty and layout generalization. These results motivated the pivot to a training platform where controlled physical cards replace uncontrolled instrument recognition.

Lessons learned. Detection worked reliably in controlled, matched-condition settings and degraded under every variation that SPD workflows present—variable illumination (mAP50-95 as low as 0.781), reflective tray surfaces (mAP50-95 as low as 0.759), and overlapping instruments (mAP50-95 = 0.389). Per-manufacturer validation is necessary because cross-brand detection drops are substantial, as shown by Kienle et al. [81]. A CV-based tray checker would therefore be viable only under specific conditions: the deployment site must provide training data covering local instruments, lighting, and tray layouts; the system needs either a closed-set operating assumption or a reliable open-set rejection mechanism; low-confidence and lookalike detections require human-in-the-loop confirmation; and the workflow must be tested in a live SPD environment. Controlled lighting and layout are prerequisites for reliable detection. The training-centered pivot is the subject of Section 6.

5.4 CV Requirements Assessment and Gap Analysis

The original tray-verification concept defined requirements for automated tray-status logic, technician-facing UI, physical capture fixture, and detection reliability. Each requirement is assessed below.

Table 10: CV design outcome versus requirements

Requirement	Target	Achievement	Gap
Tray-status logic with two-threshold confidence band	Present, Needs Review, Missing, Rescan states	Logic specified and documented	Not tested with real SPD workflow data
Technician-facing UI with one primary action per state	Clear hierarchy, fast correction, specific review reasons, human sign-off	UI requirements specified	Not implemented beyond design documentation
Physical capture fixture	Overhead camera on portable weighted board	Fixture design documented	Not fabricated or tested in SPD conditions
Lighting robustness	Detection maintains mAP50 above 0.7 across SPD-like illumination range	Evaluated across 800–80 lumens; overlay layout caused largest drop	mAP50-95=0.389 in overlay condition; lighting still a deployment barrier
Lookalike discrimination	Correctly distinguish straight/curved, narrow/broad, same-family pairs	Evaluated with 6 real instrument classes (scissor1–4, forcep, scalpel)	Per-class AP varies by split from 0.000 to 0.862; position cues dominate over shape; only visually distinct scalpel transfers reliably
Open-set rejection	Detect unknown objects as unfamiliar rather than assigning spurious labels	Not evaluated — no distractor dataset available	
Synthetic-to-real transfer	Synthetic pretraining enables few-shot real	Not evaluated — no 3D instrument models available for synthetic data generation	

The training-centered pivot (Section 6) preserved the project’s technical contributions (the CV experiments, data collection pipeline, and detection code are documented in Section 5) while replacing the main evidence path with a claim that was buildable, testable, and interpretable within the project’s resources. The CV infrastructure remains available as a future support layer after separate validation (Section 5.3).

5.5 CV Implementation Summary

The following CV subsystems were implemented:

- **CV data collection and annotation pipeline.** A webcam capture tool with bounding-box annotation UI supports rapid multi-session data collection with per-class image output, YOLO label export, and session save/load `src/trayguard/data_collection/`.
- **YOLO training infrastructure.** A Hydra-configured training pipeline supports model, dataset, and hyperparameter overrides across YOLO11n/s/m/l variants and RT-DETR-l. Training runs are deterministic and reproducible through a fixed seed `src/trayguard/training/`.
- **CV baseline: Lavado surgical instruments.** A YOLO11s model was trained on the Lavado dataset (2,106 images, 4 classes) and validated on a held-out test set, achieving mAP50 of 0.990 and mAP50-95 of 0.944 `data/lavado/`.
- **CV experiments: lighting and occlusion robustness.** Seven staged brightness experiments (800–80 lumens) on matte, reflective, separated, and overlay layouts were trained and validated. Per-stage precision, recall, mAP50, mAP50-95, and per-image count-error metrics were collected across 11 brightness levels and both backgrounds `scripts/validate_brightness_experiments.py`.
- **CV experiments: lighting diversity.** Three staged spoon lighting experiments tested whether adding lighting variety during training improves held-out-condition detection across reference, 45-degree, and 90-degree flashlight conditions `data/spoon_lighting_yolo/`.
- **Multi-variant benchmark harness.** A benchmark harness supports YOLO11n/s/m/l at 640 and 960 image sizes and RT-DETR-l at 640 for precision, recall, mAP, and latency comparison `src/trayguard/training/benchmark.py`.
- **Legacy Streamlit tray-check demo.** A live YOLO tray-check interface demonstrates camera capture, detection, and tray-status comparison for proof-of-concept review `src/trayguard/app/streamlit_app.py`.

6 Reason for the Training-Centered Pivot

(Matthew)

The project originally aimed at a live CV tray checker: detect instruments, compare detections against a tray list, and return a readiness result. That direction remains attractive, but it has the wrong proof burden for our current project. A useful clinical tray checker would need strong evidence across real instruments, local tray definitions, similar variants, glare, occlusion, unknown objects, workflow interruptions, false alerts, human review, device policy, cleaning, privacy, and adoption. The current project can demonstrate parts of that pipeline, but does not claim deployment-grade clinical reliability. The Week 6 pivot made the value proposition more testable: reduce simulated time-to-competency and training burden for sterile-processing learners by building an interactive training platform. That direction keeps the same root problem, because missing, wrong, extra, and misidentified instruments still arise during tray reconstruction. It changes the claim from "the model can outperform humans in a hospital environment" to "novices can improve on a measured local tray simulation after retrieval-centered practice." The pivot matches the evidence: sterile-processing work depends on local knowledge and supervised practice, and simulation/retrieval evidence supports repeated practice with feedback. It is buildable with low-cost materials: a file-backed tray module, printable cards, a scanning surface,

scoring, and export. It also reduces risk for a future CV system by creating validated local content, error categories, workflow language, and learner data before asking a camera model to make clinical-grade distinctions.

7 Final Product Concept And Design Rationale (Matthew, Andy, Qiyu)

7.1 Final Product Concept

TrayGuard is a local tray training and assessment platform. The core product is a learner-facing app plus a file-backed tray module that defines the instruments, aliases, required counts, distractors, lookalike pairs, distinguishing features, and assessment variants for one local tray. The first build should use printable AprilTag instrument cards as the physical simulation layer. Each card represents one instrument or instrument variant. The app scans the cards on a tabletop or tray area, treats the detected markers as the learner's tray selection, scores the result against the tray template, and returns missing, extra, wrong, misidentified, and wrong-count feedback in practice mode. Pre-test and post-test modes suppress hints and corrective feedback so the same system can collect baseline and post-training evidence. The full learning loop is:

```
local tray module -> pre-test -> retrieval-first cards -> quiz -> AprilTag-card tray sort -> post-test ->
metrics export
```

This concept keeps the prototype buildable while preserving the important task structure: the learner must use local names, local counts, visual/semantic distinctions, and tray rules to assemble a simulated tray.

7.2 Alternatives Considered

Several product directions were considered:

Table 11: Alternative product directions considered and rejected

Alternative	Benefit	Reason rejected or deferred
Deployment-grade CV tray checker on real instruments	Closest to the original automation concept and strongest if validated in an SPD.	Requires local instrument data, clinical workflow integration, failure-mode validation, and regulatory/adoption work beyond our project timeline.
Software-only study and quiz module	Fastest implementation path and easiest novice study.	Too far from the tray reconstruction task; it would mostly test recognition, not applied count-sheet use.
Real teaching instruments with camera-supported tray review	Better physical fidelity than cards.	Requires access to representative instruments, cleaning/handling rules, storage, and educator review; appropriate for an SPD pilot after the simulated loop works.
Printable QR/AprilTag instrument cards	Cheap, portable, repeatable, and compatible with device scanning; closely matches NeuroVase-style tangible cue-card learning.	Less physically realistic than real instruments, so claims must stay limited to simulated tray-task improvement.
Manual drag-and-drop tray simulation	Simple software implementation without camera setup.	Does not exercise physical search, placement, or scanning workflow, and is less persuasive for an interactive project demo.

The selected first prototype is the printable AprilTag-card path because it creates a physical, measurable simulation without depending on deployment-grade instrument recognition.

7.3 Selected Candidate and Backup Paths

The selected candidate is:

A local tray training and assessment app that uses printable AprilTag instrument cards for simulated tray sorting, retrieval-first study, quiz, practice feedback, no-hints pre/post assessment, and metrics export.

The backup paths are layered so the project still produces evidence if one piece fails.

Table 12: Backup path strategy by risk level

Level	Selected path	Backup path	Claim preserved
Learning feature	Retrieval cards, quiz, practice sort, pre/post sort.	If the full loop is too long, run pre-test, study cards, practice sort, and post-test only.	Immediate simulated learning.
Physical interface	AprilTag cards scanned on a tray mat.	Use QR codes or manual card selection if AprilTag scanning is unreliable.	Local tray-task learning, with marker reliability reported as a limitation.
Content	<code>basic_general_tray_v1</code> with 8-12 required concepts, distractors, lookalikes, and matched variants.	Reduce to 6-8 required concepts while preserving at least two lookalike pairs and count errors.	Bounded module feasibility.
CV	Marker detection only for project scoring.	Defer all camera recognition and score user selections manually.	Training evidence without CV overclaim.
Evaluation	8-12 novice participants plus delayed retention if schedule allows.	Run a smaller formative walkthrough and report it as usability/protocol evidence, not learning proof.	Risk-reduction handoff.
Content validity	SPD educator or instructor review.	Faculty or surgical-technology reviewer plus explicit future SPD review requirement.	Plausibility review, not clinical validation.

This backup strategy is intentionally conservative. It protects the project from collapsing into a fragile detector demo and keeps the main deliverable aligned with the training pivot.

7.4 Rationale for AprilTag Marker Choice

The prototype uses printable AprilTag fiducial markers on physical instrument cards. This section justifies AprilTag specifically against the alternatives of no physical component, QR codes, ArUco markers, and real instruments with camera-based recognition.

7.4.1 Why Physical Markers at All

A purely digital drag-and-drop tray simulation would be simpler to implement and would avoid camera-setup and detection-reliability concerns. It is rejected because the target task is fundamentally physical: SPD technicians search for, pick up, examine, count, and arrange real instruments under a tray boundary [3]; [1]. A purely on-screen interface removes the spatial search, the need to distinguish similar objects in hand, and the physical count-and-place workflow that the study is meant to simulate. Physical cards preserve those cognitive demands while replacing regulated stainless-steel instruments with safe, cheap, tractable tokens.

7.4.2 Why AprilTag over QR Codes, ArUco, or Real Instruments

AprilTag was chosen over QR codes, ArUco markers, and real instruments for reasons of detection robustness, error correction, and study feasibility. Compared with QR codes, AprilTag provides faster corner detection at sub-pixel precision, a lexicode-based Hamming-distance guarantee that corrects bit

errors from glare or occlusion, and native pose estimation for future spatial feedback [89]; [90]. Compared with ArUco, AprilTag’s lexicode dictionary guarantees a larger minimum inter-marker distance, and its corner-refinement pipeline produces more stable localization across lighting and image scales [91]. Compared with real instruments — the closest alternative to the original CV-checker goal — physical cards avoid the regulatory, handling, and recognition burdens that would dominate the engineering schedule, and follow the same abstraction strategy as NeuroVase’s tangible cue cards [70]; [81]. The recommended tag family is `tagStandard52h13` (52 data bits, minimum Hamming distance 13), supported through OpenCV’s `aruco` module with no additional library dependency [92].

7.5 Rationale for Gamified and Simulation-Based Learning Style

TrayGuard embeds retrieval practice, feedback, scoring, progressive difficulty, and confidence capture within a structured learning session. Simulation-based medical education with deliberate practice has been shown to outperform traditional clinical education, and gamification combining assessment features with progressive challenge increases engagement and learning outcomes [66]; [93]. Table 13 maps each feature to its evidence base.

Table 13: Game elements and their learning-science rationale

Game-like feature	Learning mechanism	Evidence base
Pre-test establishes a baseline score	Goal-setting and self-assessment calibration before practice	Pre-testing improves later retention compared with studying without a prior test, even when answers are not provided [86]; [94].
Retrieval cards use prompt-before-reveal	Active recall with immediate confirmation; the core retrieval-practice effect	Tests improve long-term retention more than restudy across health-professions contexts [95]; [69].
Quiz with score and weak-item logging	Retrieval with spaced repetition and targeted remediation	Weak-item repetition reduces error persistence on later tasks [24]; [96].
Practice tray sort with error-category feedback	Deliberate practice with specific, actionable correction	Immediate error-specific feedback improves skill acquisition more than correctness-only feedback [96]; [68].
Confidence capture after each item	Metacognitive calibration; exposes high-confidence errors	High-confidence errors are more resistant to correction and indicate overconfidence that feedback must target explicitly [24].
Parallel no-hints post-test	Near-transfer assessment without answer leakage	Separating assessment from practice prevents score inflation from memorization of exact layouts [97].
Metrics export and weak-item summary	Learner- and instructor-facing performance data for targeted remediation	Instructors cannot target weak items or weak rules without item-level evidence [26]; [3].

The design deliberately avoids leaderboards, badges, and competitive ranking, which can shift motivation from intrinsic learning to extrinsic reward-seeking [93]. Full VR simulation (e.g., SteriBoost) and narrative-driven modules (e.g., Touch Surgery) were considered but rejected because they add hardware or authoring burden without increasing the number of tray-repetition cycles per session [98]; [78]. The chosen level is closest to the serious-game approach of SteriDefi—a question-bank and scoring system that motivates repeated knowledge retrieval [99].

8 Requirements Definition

(Matthew, Andy, Jacob)

8.1 Element Definition

Table 14: Element definition for the TrayGuard system

Element	Definition For TrayGuard
Intended practice	Novice preparation for local tray reconstruction before or alongside supervised SPD training.
Other practice affected	Instructor review, clinical supervisor coaching, tray content updates, weak-item remediation, and future CV-supported tray review.
Artifact	A local training app, file-backed tray module, printable marker cards, tray-sorting surface, scoring engine, and export/report package.
Problem addressed	Learners need repeated, measurable practice applying local tray rules, counts, aliases, and lookalike discrimination before real tray work.
Technology	Local web/app workflow, JSON tray modules, QR or AprilTag cards, camera or manual selection input, scoring logic, and CSV/JSON export.
Primary users	Novice learners, instructors, SPD educators, and future project teams.
Perception goal	The system should feel like a serious practice tool, not a clinical approval device or punitive surveillance system.
Environment	Classroom, skills lab, project demo, or supervised training setting; not a live sterile field or production SPD deployment.
Core function	Convert a local tray module into study, quiz, simulated sorting, feedback, assessment, and learner metrics.
Required behavior	Separate practice from assessment, suppress hints during pre/post modes, classify errors consistently, and export comparable rows.
Structure	Content module, learning workflow, marker/manual input, scoring engine, feedback engine, reporting/export, and instructor review.
Intended effects	Improve simulated tray-sorting accuracy, reduce severe errors, expose weak items, calibrate confidence, and reduce early clinical supervisor burden.
Possible side effects	Overclaiming simulated scores, punitive use of learner metrics, answer leakage, inequitable access to updated modules, and reduced attention to supervised hands-on practice.

8.2 Objective Tree

Improve novice readiness for local tray reconstruction through repeatable, measurable simulated practice.

1. Increase local tray knowledge

1.1 Teach local names, aliases, and instrument families

- 1.2 Teach required counts and tray-specific rules
- 1.3 Teach lookalike and same-family distinctions
- 2. Improve applied tray-sorting performance
 - 2.1 Reduce missing required items
 - 2.2 Reduce wrong and misidentified items
 - 2.3 Reduce extra and wrong-count items
 - 2.4 Maintain reasonable task time without rewarding guessing
- 3. Make learning measurable
 - 3.1 Capture pre-test and post-test accuracy
 - 3.2 Export item-level error categories
 - 3.3 Capture confidence, Not sure choices, and high-confidence errors
 - 3.4 Preserve comparable assessment variants
- 4. Support instructors
 - 4.1 Show weak instruments and weak tray rules
 - 4.2 Keep modules local, versioned, and reviewable
 - 4.3 Avoid collecting patient data or unnecessary learner identifiers
- 5. Preserve claim boundaries
 - 5.1 Label outputs as simulated practice evidence
 - 5.2 Require expert review before content-valid claims
 - 5.3 Require SPD pilot evidence before workplace-effectiveness claims

8.3 Performance Specifications

The first build should use the following targets. They are deliberately formative rather than clinical: they define when the prototype is ready to support a project novice-learning claim.

Table 15: Performance specifications and acceptance targets

Area	Target	Evidence Basis	Measurement
Participant count	Minimum 8 novice participants; target 12 complete participants if schedule allows.	NeuroVase used a 40-participant controlled study for an educational card/AR system, while qualitative methods literature supports small, narrow, homogeneous samples for early formative work [70]; [100]; [101].	Unique participants with complete pre/post attempts.
Completion rate	At least 7 of 8 participants, or 85% when $n > 8$, complete the full loop without blocking moderator intervention.	Usability must be separated from learning; NeuroVase used SUS and user-experience measures alongside knowledge testing.	Completion flag and blocking help requests.
Accuracy gain	Mean paired tray accuracy improves by at least 20 percentage points from pre-test to post-test.	Ofstead's SP training pilot showed large pre/post score movement after structured teaching and hands-on practice; TrayGuard should set a smaller but visible formative threshold [24].	<code>post_accuracy - pre_accuracy</code> .
Post-test floor	Mean post-test accuracy is at least 80%, and no more than one participant scores below 70%.	The goal is not only improvement from a low baseline; learners should reach a usable simulated familiarity level before the module is considered ready.	Post-test <code>accuracy_score</code> .
Severe-error reduction	Mean <code>missing + wrong + misidentified</code> errors decrease by at least 30%.	Alfred et al. and Nichol et al. show missing, wrong, extra, and assembly-related errors are operationally meaningful categories [1]; [3].	Paired error-category counts.
Speed without unsafe guessing	Median post-test duration is at least 10% lower, or is no more than 20% higher if accuracy improves by at least 20 points.	Time matters because tray assembly is workflow-bound, but speed alone could reward guessing.	<code>duration_seconds</code> , paired by learner.
Confidence calibration	High-confidence errors decline by at least 30%; low-confidence correct answers are exported.	Ofstead captured confidence/satisfaction, and feedback research supports showing learners where they are and what to do next [24]; [96].	Confidence ≥ 4 on incorrect items; confidence ≤ 2 on correct items.
Retention	A 7-day delayed tray sort retains at least half of the immediate accuracy gain 29 and delayed accuracy does not return to within 5 percentage points of the	Bell et al. showed immediate medical-learning gains can decay within days and recommended reinforcement after as	Delayed <code>accuracy_score</code> relative to pre/post.

8.4 Quality Function Deployment

Compact QFD:

Table 16: Quality function deployment: needs to engineering targets

Stakeholder need	Engineering characteristic	Target value	Verification
Learners can practice local tray content repeatedly.	File-backed module with required items, distractors, aliases, features, counts, and variants.	One complete <code>basic_general_tray_v1</code> module with 8-12 required concepts and 3-6 distractors.	Module schema review and demo load.
Practice feels connected to tray reconstruction, not isolated memorization.	Simulated tray sort using physical cards or manual selection.	Pre-test, practice, post-test, and optional delayed variant all use the same tray rule.	End-to-end run.
Educators can see what learners missed.	Attempt-level and item-level export.	100% of completed runs export all required fields.	Schema validation.
Errors map to operationally meaningful categories.	Scoring engine with <code>missing</code> , <code>extra</code> , <code>wrong</code> , <code>misidentified</code> , and <code>wrong_count</code> .	Every submitted tray receives category counts.	Unit tests or scored example cases.
Assessment is not contaminated by coaching.	Mode boundaries and feedback suppression.	No hints or corrective feedback in pre-test, post-test, or delayed modes.	UI walkthrough and export mode check.
The physical marker layer does not corrupt scores.	Marker reliability and user confirmation.	At least 95% detection in normal layouts; confirmation before scoring.	Study 0 marker log.
Learner confidence is visible.	Confidence or <code>Not sure</code> capture.	High-confidence errors and low-confidence correct answers exported.	Export inspection.
The system remains ethical and nonpunitive.	Privacy and report labels.	Hashed learner IDs, no patient data, and report language that says practice evidence.	Report review.
Future teams can extend the design.	Versioned data model and traceability.	Module version, source notes, verification status, and assessment variant IDs included.	Data file review.

8.5 Function Analysis – Transparent Box Model

The training system is modeled as a transparent box that transforms a local tray module and physical instrument cards into learner performance evidence through sequential subfunctions [104]:

Table 17: Transparent box model for the training system

Input	Subfunction	Output
Local tray module (JSON)	Authoring and verification	Verified tray template with required items, distractors, lookalike pairs, and assessment variants
Verified tray template	Pre-test assessment	Baseline accuracy, error-category counts, and confidence distribution
Instrument cards and tray template	Retrieval-first study	Prompt-before-reveal card practice with distinguishing features and aliases
Study session output	Identification quiz	Quiz accuracy, duration, and item-level confidence per instrument concept
Cards and tray template	Simulated tray sorting	Selected instrument set scored against required counts
Scored tray selection	Feedback generation	Missing, extra, wrong, misidentified, and wrong-count error report
Pre-test and post-test scores	Metrics aggregation	Pre/post accuracy gain, error reduction, confidence calibration, and weak-item report
Aggregated metrics	Export and reporting	CSV/JSON export with attempt-level and item-level records

8.6 Morphological Chart

The morphological chart generates solution alternatives for each subfunction identified in the transparent box model [105, 104].

Table 18: Morphological chart: solution alternatives by subfunction

Subfunction	Alternative 1	Alternative 2	Alternative 3	Alternative 4
Content delivery	File-backed JSON module	Online database	Spreadsheet import	Hard-coded tray lists
Instrument representation	Physical marker cards	On-screen instrument icons	Real instruments	Text-based item list
Marker technology	AprilTag	QR code	ArUco	Manual selection
Learner interaction	Study + quiz + tray sort	Study-only	Quiz-only	Tray-sort only
Assessment mode	Pre/post with matched variants	Single post-test	Pre/post without variants	Continuous assessment
Feedback mechanism	Error-category report	Pass/fail only	Item-level hints	Instructor review only
Scoring approach	Category-based scoring (missing, extra, wrong, misidentified, wrong-count)	Simple accuracy (correct/total)	Weighted error severity	Confidence-adjusted scoring
Export format	CSV + JSON per run	Single PDF report	Instructor dashboard	Raw database access

The selected combination is: file-backed JSON module, physical AprilTag marker cards, full study-quiz-sort learning loop, pre/post matched-variant assessment, category-based scoring, error-category feedback, and CSV/JSON export.

8.7 Weighted Objective Method

The weighted objective method scores each candidate solution combination against weighted criteria derived from the objective tree and stakeholder needs [105].

Table 19: Weighted objective method: solution path selection

Criterion	Weight	Training path	CV tray checker	Software-only quiz	Real-instrument CV
Buildability within project scope	0.25	5	1	4	1
Measurable learning evidence	0.20	5	2	3	3
Physical task fidelity	0.15	3	5	1	5
Low validation burden	0.15	4	1	4	1
Instructor usefulness	0.10	4	3	3	3
Future extensibility	0.10	3	4	2	4
Cost and equipment requirements	0.05	5	3	5	2
Weighted total	1.00	4.35	2.40	3.15	2.60

The training path scores highest (4.35/5.00), driven by buildability, measurable evidence, and low validation burden.

8.8 Requirement Traceability Matrix

Table 20: Requirement traceability matrix

Req.	Stakeholder need	Design feature	Evidence source	Validation method	Status
FR-1 Load local module	Educators need updateable local tray rules.	JSON tray module and module selector.	Count-sheet and local tray evidence [23].	Change module data and reload.	Ready for impl.
FR-2 Retrieval cards	Learners need active recall of names, features, and counts.	Prompt-before-reveal cards.	Retrieval and flashcard evidence [86]; [69].	Card walkthrough and quiz logs.	Ready for impl.
FR-3 Pre/post assessment	Reviewers need measurable learning evidence.	No-hints pre-test and post-test variants.	Ofstead pre/post pattern [24].	Paired run export.	Protocol-ready.
FR-4 Quiz mode	Learners need feedback before applied sorting.	Quiz with confidence and weak-item logging.	Feedback and retrieval evidence [96].	Quiz result export.	Ready for impl.
FR-5 Practice feedback	Learners need actionable correction.	Error-specific feedback engine.	Tray defect categories [1]; [3].	Seeded error cases.	Defined.
FR-6 Metrics export	Instructors need reviewable evidence.	Attempt and item CSV/JSON.	Error-reporting and competency evidence [3]; [24].	Schema validation.	Defined.
FR-7 Mode separation	Assessment must remain fair.	Mode state and feedback suppression.	Assessment design and answer-leakage risk.	UI walkthrough.	Defined.
NFR-1 Fast data load	Demo cannot stall.	Local file storage.	Usability expectations.	Load timing ≤ 2 seconds.	Future test.
NFR-2 Short flow	Participants can finish in one session.	Bounded module and timed segments.	Study protocol and sample feasibility.	Timing ≤ 20 min for demo.	Future test.
NFR-3 Export reliability	No completed run should be lost.	Local export after each run.	Study evidence requirement.	100% completed-run exports.	Future test.
NFR-4 Privacy	Learners should not be surveilled unnecessarily.	Hashed IDs and no PHI.	Healthcare data governance concerns.	Export review.	Defined.
NFR-5 Content traceability	Local content must be auditable.	Source notes and instructor verification status.	Count-sheet and local content boundary.	Module review.	Defined.
NFR-6 Capability labels	Stakeholders must not overread scores.	Report language: simulated practice evidence.	Certification and clinical-transfer boundaries.	Report review.	Defined.

9 System Design

(Matthew, Jacob)

9.1 System Overview

TrayGuard should be built as a local module runner plus a card-scanning assessment surface. The app does not need to recognize real instruments for the first study. It needs to observe which simulated instrument cards the learner placed in the tray area, score that selection against a local tray template, and export evidence.

```
local tray module
-> pre-test AprilTag tray sort
-> retrieval-first study cards
-> quiz with confidence / Not sure
-> practice AprilTag tray sort with feedback
-> post-test AprilTag tray sort
-> metrics export
-> committed 7-day delayed retention sort (see falsification criteria in Section~\ref{sec:study-2})
```

The architecture follows the evidence base:

- Local tray modules are required because count sheets define tray names, contents, quantities, sizes, and reference/catalog numbers, and real count sheets include fields such as reference number, description, location, quantity, check box, total count, and hospital signature [23]; [62]; [63].
- Retrieval-first study and quiz modes are justified by retrieval practice, health-professions flashcard evidence, and spaced digital education [86]; [69]; [106].
- Pre/post practice, confidence capture, and optional delayed retention follow the sterile-processing precedent in Ofstead and the retention warning in Bell et al. [24]; [97].
- AprilTag cards are a physical proxy, not a real-instrument detector. The tangible-card idea is supported by NeuroVase-style cue cards and board/card-based healthcare education, but clinical transfer remains future validation [70]; [74].

9.2 Subsystem Breakdown

The system separates into these functional subsystems:

Table 21: Functional subsystem breakdown

Subsystem	Responsibility
Local content/tray module	Stores instrument records, aliases, photos, distinguishing features, required quantities, distractors, and tray-template versions.
Learning workflow	Routes the learner through pre-test, study, quiz, practice sort, weak-item review, post-test, and export.
Retrieval practice	Presents prompt-before-answer cards and quizzes for instrument names, features, lookalikes, and required counts.
AprilTag card scanning	Detects printed marker cards and maps marker IDs to instrument IDs so the app can observe a physical tray simulation.
Simulated tray scoring	Compares detected cards and selected quantities against the tray template.
Feedback engine	Reports missing, extra, wrong, misidentified, and wrong-count errors in practice mode.
Confidence/uncertainty capture	Records confidence or Not sure responses for assessment and instructor review.
Reporting/export	Produces pre/post accuracy, time, error-category, confidence, and weak-item summaries.
Instructor review	Lets an educator verify tray content, local naming, distractors, and study evidence before broader use.

Computer vision for recognizing real instruments remains a future support subsystem. In the project build, the marker-scanning subsystem is a proxy for observing user decisions, not an instrument-identification system.

9.3 Data Model and Module Contract

The data model should mirror a real count sheet while staying small enough for the first build. Stryker’s public count sheets show the practical minimum: reference number, description, location/layer, quantity, check box, total count, additional items, and hospital signature. TrayGuard adds learning-specific fields for aliases, lookalikes, confidence, and error categories.

Table 22: Data model objects and required fields

Object	Required Fields	Why It Matters
Instrument	id, display_name, family, aliases, distinguishing_features, image_refs, apritag_id, local_verification_status	Supports retrieval cards, quiz prompts, card scanning, and local naming.
TrayTemplate	id, name, version, procedure_family, source_note, required_items, distractor_items, lookalike_pairs, assessment_variants	Stores the local count-sheet task and separates required instruments from distractors.
TrayItemRequirement	instrument_id, quantity, reference_number, location_or_layer, acceptable_substitutes, notes	Mirrors real count-sheet fields and makes count/location errors measurable.
AssessmentVariant	id, mode, random_seed, required_item_ids, distractor_item_ids, card_front_policy, feedback_enabled	Keeps pre/post/delayed tasks comparable while preventing answer leakage.
LearningRun	run_id, learner_id_hash, participant_group, tray_template_id, variant_id, mode, started_at, completed_at	Separates study, quiz, practice, pre-test, post-test, and delayed attempts.
DetectedCard	run_id, apritag_id, instrument_id, timestamp, detection_confidence, confirmed_by_user	Lets marker failures be audited instead of silently counted as learner errors.
AttemptResult	run_id, selected_items, duration_seconds, accuracy_score, required_recall, errors, confidence_summary	Provides the main pre/post evidence.
ItemResult	run_id, instrument_id, required_qty, selected_qty, error_category, confidence, not_sure	Supports item-level feedback, weak-item review, high-confidence error reporting, and instructor review.

The first implementation can store these objects in JSON files and export CSV for analysis. The important constraint is schema stability: pre-test, post-test, and delayed retention must produce comparable rows. Sample instrument JSON:

```
{
  "id": "mayo_scissors_straight_55",
  "display_name": "Straight Mayo scissors, 5.5 in",
  "family": "cutting_dissecting",
  "aliases": [
    "straight Mayo",
    "suture scissors"
  ],
  "distinguishing_features": [
```

```

    "Heavy straight blades",
    "Broader blade profile than Metzenbaum scissors",
    "Ring-handled cutting instrument"
  ],
  "common_confusions": [
    "mayo_scissors_curved_55",
    "metzenbaum_scissors_curved_55"
  ],
  "image_refs": [
    {
      "id": "view_a",
      "path": "data/instruments/mayo_scissors_straight_55_view_a.jpg",
      "source": "local_photo",
      "approved_for_study": true,
      "approved_for_assessment": true,
      "attribution": "Local TrayGuard photo"
    },
    {
      "id": "card_front_neutral",
      "path": "outputs/cards/mayo_scissors_straight_55_neutral_front.png",
      "source": "generated_card",
      "approved_for_study": false,
      "approved_for_assessment": true,
      "attribution": "TrayGuard AprilTag card"
    }
  ],
  "apriltag_id": 12,
  "card_front_policy": "neutral_assessment_front",
  "study_prompts": [
    {
      "id": "name_recall",
      "prompt": "Name this instrument.",
      "answer": "Straight Mayo scissors, 5.5 in"
    },
    {
      "id": "feature_recall",
      "prompt": "What feature helps distinguish this from Metzenbaum scissors?",
      "answer": "Straight Mayo scissors have heavier, broader blades."
    }
  ],
  "notes": "Use local photos before pilot. Do not rely on vendor images for assessed study content.",
  "local_verification_status": "pending_review"
}

```

9.4 Mode Behavior

Table 23: Training mode behavior by hints, feedback, and metrics

Mode	Hints	Corrective Feedback	Metrics	Use
Pre-test	No	No	Yes	Baseline simulated tray-task proficiency.
Study cards	Yes	Yes	Optional	Prompt-before-reveal retrieval of names, aliases, families, features, and counts.
Quiz	Limited after answer	Yes	Yes	Retrieval practice, confidence calibration, and weak-item discovery.
Practice sort	Yes	Yes	Yes	Applied retrieval and deliberate practice on tray organization.
Post-test	No	No	Yes	Comparable same-day assessment after practice.
Delayed retention	No	No	Yes	Checks whether gains persist after 7 days (with falsification criteria).

9.5 Learning Loop

The learning loop (Section 7) applies retrieval practice as the instructional engine. The learner should usually commit to an answer before the system reveals it: name the instrument, identify a distinguishing feature, choose the lookalike, recall the count, or decide whether the item belongs in the tray. The simulated tray sort is applied retrieval: the learner must use those same facts in context with distractors and quantities.

Each feature in the learning loop maps to a specific evidence-based role:

Table 24: Feature-to-evidence mapping for the learning loop

Feature	Evidence-based role	Engineering requirement
Retrieval-first study cards	Initial instruction that still asks the learner to answer before reveal	Store photos, names, aliases, functions, distinguishing features, and local notes; present prompts before explanations
Identification quiz	Central retrieval practice for instrument identity and features	Ask recall or recognition questions, randomize distractors, record correctness, confidence, and time
Simulated tray sorting	Safe applied retrieval in a realistic work task	Let learners assemble or check a tray against a count sheet in a low-risk training environment
Immediate error-specific feedback	Deliberate practice and error correction	Explain wrong answers and classify errors as missing, extra, wrong, misidentified, or wrong count
Confidence ratings	Metacognitive signal for learner and instructor	Capture confidence before or after tasks so low-confidence correct answers and high-confidence errors are visible
Separated pre/post tests	Evaluation not contaminated by coaching	Disable hints and feedback during tests; export comparable pre/post accuracy, time, error type, and confidence

9.6 Module Scope

The first module should contain:

- 8-12 required instrument concepts;
- 3-6 distractor instruments;
- 2-4 known lookalike or same-family pairs;
- local names, aliases, family labels, required counts, and distinguishing features;
- AprilTag cards that represent instruments during the physical tray simulation;
- parallel pre-test, post-test, and delayed-retention tray variants with matched difficulty;
- instructor or expert review before results are presented as content-valid.

The cards should use neutral assessment fronts. Names, counts, and obvious answer labels should be hidden during pre-test, post-test, and retention tasks so the participant cannot solve the task by reading the card.

9.7 FGVC-Derived Module Plausibility

The FGVC-derived modules are practice modules built from the major-tray image classes reported by Atabuzzaman et al. [61]. The source contributes instrument names and images for a bounded image-card set: scissors, clamps, forceps, needle holders, retractors, suction, and other classes present in the major-tray archive. It gives TrayGuard a coherent pool of real instrument classes photographed under one dataset program.

Plausibility for these modules comes from three linked sources of evidence. First, every required card, distractor card, and lookalike pair is selected from the FGVC major-tray class mapping. Second, the module schema mirrors real tray/count-sheet fields: tray name, required item, quantity, reference number, location or layer, total count, and check status [62]; [63]. Third, the module size follows the project study constraint: 8-12 required concepts, 3-6 distractors, and 2-4 lookalike pairs, so novices can complete pre-test, practice, post-test, and delayed retention tasks in a single session. The FGVC-derived quantities are simulated as one card per selected class. Expert review sets the final clinical wording, counts, substitutes, and local tray membership before an SPD pilot uses the module as content-valid tray material.

9.8 Real Tray List Examples

The first TrayGuard module should be smaller than a clinical tray, but its data fields should look like real tray/count-sheet data. Three examples are useful:

Table 25: Real tray list examples and design implications

Real Example	What It Shows	Design Implication
Stryker Gamma4 Indication Tray count sheet	A real vendor count sheet uses tray name, insert/base sections, reference numbers, descriptions, locations, quantities, check boxes, total instrument counts, additional items, and hospital signature [62].	TrayGuard should store reference_number , description/display_name , location_or_layer , quantity , and additional_items fields even if the first module only uses names and quantities.
Stryker IMN Basic Instruments Tray count sheet	A second real count sheet uses the same pattern and splits the tray into insert/base groupings with total counts of 15 and 10 instruments [63].	The data model should support sections/layers and total-count validation, not just a flat list.
Multi-specialty tray rationalization project	Rubak et al. consolidated, separated, or restructured 1,340 different instrument trays across surgical specialties at one university hospital, finding that different specialties required different tray structures [64].	Real tray lists scale to thousands of configurations in a single hospital. The data model must be open-ended beyond the first study's 8-12 required concepts.
Published major orthopedic tray study	Toor et al. observed an 88-instrument major orthopedic tray across 80 procedures and compared optimized tray configurations of 47, 67, and 51 instruments [107].	Real clinical trays can be far larger than the first module. The first study should use 8-12 required concepts for feasibility, while preserving fields that can scale to larger trays.

For the first build, use a teaching-scale tray: 8-12 required concepts, 3-6 distractors, and 2-4 lookalike pairs. This is intentionally smaller than the real examples so a novice can complete the loop in 45-60 minutes.

10 Analysis

(Matthew, Qiyu)

10.1 Prototype Capability Demonstration

The following capabilities have been demonstrated in the built prototype:

Table 26: Demonstrated prototype capabilities

Capability	Evidence	Status
Tray module loading	<code>basic_general_tray_v1</code> module loads and renders in the web app	Verified
Retrieval-first study cards	Prompt-before-reveal cards display instrument names, images, aliases, features, and tray quantities	Verified
Identification quiz	Quiz presents instrument images and captures name, confidence, and Not sure responses	Verified
AprilTag card detection	Camera-based detection maps marker IDs to instrument IDs in real time; manual entry fallback available	Verified
Simulated tray scoring	Submitted selections scored against tray template with accuracy and error categories	Verified
Error-category classification	Missing, extra, wrong, misidentified, and wrong-count errors computed per run	Verified
Pre/post assessment mode	No-hints mode with matched variants; baseline and post-training evidence collected	Verified
Confidence capture	1–5 scale and Not sure checkbox recorded per item; high-confidence errors flagged	Verified
CSV/JSON export	Attempt-level and item-level records exported with all required fields	Verified
Card generation	CLI generates printable marker PNGs, grid index sheets, and double-sided study flashcards	Verified

Every verified capability supports a direct link to the performance specifications (Section 8.3) and the requirements traceability matrix (Section 8).

10.2 Marker Reliability Analysis

The marker layer is a measurement tool: if detection fails, scores confound learner error with app error. The following analysis frames the reliability expectation:

Table 27: Marker reliability analysis

Dimension	Target	Method
Detection rate in normal layouts	$\geq 95\%$	Bench test with varied card spacing, rotation, and lighting
Detection rate in degraded conditions	Reported as limitation	Glare, partial occlusion, and extreme angles logged per test
Marker-ID confusion rate	0% in controlled test	All markers in the printed set decoded against ground truth
False positive rate (non-marker objects)	Reported as limitation	Objects placed in camera view that do not contain markers
User confirmation before scoring	Required	Confirmation screen shows detected cards for user review before scoring

The target values match the performance specification (Section 8.3) and the QFD requirement for marker reliability. Study 0 in the evaluation plan (Section 11.5) provides the experimental protocol for formal measurement.

10.3 Unit Test Results

Unit tests verify core logic that directly affects learner evidence quality:

- **Module loading.** JSON schema validation confirms that all required fields are present and correctly typed for the primary tray module.
- **Scoring accuracy.** Seeded tray submissions with known error patterns (one missing, one extra, one lookalike substitution, one wrong-count) return the expected category counts and overall accuracy score.
- **Export completeness.** CSV and JSON exports from scored runs contain all required fields at both the attempt level and item level.
- **Card generation.** Generated AprilTag PNGs decode to the correct marker IDs, and the double-sided flashcard layout renders instrument data without text overflow.

10.4 Design Traceability

Every design decision in the training path traces to an objective in the objective tree (Section 8.2), a specification in the performance specification analysis (Section 8.3), and a requirement in the requirement traceability matrix (Section 8). The research support audit (Appendix A) provides choice-by-choice traceability to published literature, and the implementation defaults (Appendix B) document the critique-ready interpretation of each current implementation decision.

11 Evaluation And Validation Plan

(Matthew, Jacob)

11.1 Evaluation Framework

Two frameworks define the evaluation scope: Kirkpatrick’s four-level model and Cook and Beckman’s five-source validity framework [108, 103]. A pre/post accuracy gain on a simulated tray task is Level 2 evidence. A pre/post accuracy gain on a simulated tray task is Level 2 evidence. The current study ladder addresses Level 2 directly and builds toward Level 3 through convergent evidence and expert

review, but does not claim Level 4 (reduced OR delays or hospital defect rates), which requires an SPD-site study [72].

Applied to TrayGuard, Cook and Beckman’s validity framework requires the following evidence chain [103]:

1. **Content validity:** the tray module, names, aliases, counts, distractors, and lookalike pairs are plausible for a training tray. Addressed by expert review (Study 3) and instructor-verified module content.
2. **Response process:** learners interact with the system as intended; confidence ratings and **Not sure** capture reflect genuine self-assessment rather than confusion or guessing. Addressed by usability observation and confidence-calibration analysis (Studies 1-2).
3. **Internal structure:** error categories behave consistently: lookalike pairs produce misidentified errors, distractors produce extra errors, and repeated practice reduces specific error types. Addressed by pre/post error-type analysis (Studies 1-2).
4. **Relations to other variables:** TrayGuard tray-sort scores correlate with an independent measure of the same knowledge. This is the weakest link in the current design. Addressed by a convergent photo-identification test added to Study 1.
5. **Consequences:** the scores support the intended interpretation (improved simulated tray familiarity) and do not support unintended interpretations (clinical competence, SPD readiness). Addressed by scope boundaries and report labeling (Section 12).

A sixth issue is **population generalizability**. Novices start near zero accuracy, while SPD workers may already be proficient on familiar trays. Three responses apply: (i) Ofstead et al. show experienced SP workers are not at ceiling on novel tasks (mean pre-test 41% on borescope inspection) [24], so TrayGuard’s value for workers is error-type reduction, confidence calibration, and retention maintenance rather than absolute accuracy gain; (ii) the dependent variable shifts from pre/post accuracy gain (novices) to error-type reduction, confidence calibration, and new-module learning rate (workers), as Study 4 must specify [108]; and (iii) the learning mechanism (retrieval practice, feedback, spaced repetition) is population-independent even if the effect size is not [72, 86, 95, 94]. The study ladder (Table 28) maps each study level to the evidence source it earns; no single study provides all sources, and the claim is supported by the accumulated pattern.

11.2 Current Evidence Available

The current evidence package includes:

- literature-backed problem definition for SPD tray reconstruction and assembly errors;
- support for simulation, retrieval practice, flashcards, feedback, tangible card interfaces, and board/card-based healthcare education;
- a bounded final product concept based on local tray modules and AprilTag instrument cards, with the evaluation framework above clarifying which claims each evidence source supports;
- a subsystem architecture, data contract, mode behavior, and scoring rubric;
- a user-study plan that addresses content validity, response process, internal structure, relations to other variables through convergent evidence, and retention through a committed delayed check.

11.3 Formative Novice Study Plan

The first feasible study is a formative novice study with students or other participants who do not have SPD tray experience. The study should evaluate whether the AprilTag-card simulation teaches a bounded local tray module, not whether participants are ready for real SPD work. Use one tray module with 8-12 required instruments, a distractor pool, and parallel pre/post tray variants. A reasonable first sample is 8-12 participants for classroom evidence; a larger follow-up can use paired statistics once the prototype and protocol are stable. Study sequence:

1. Consent and prior-experience questionnaire.
2. No-hints pre-test: assemble the simulated tray from AprilTag instrument cards using a tray list or count-sheet-style prompt.
3. Retrieval-first study cards for instrument names, aliases, distinguishing features, and counts.
4. Short quiz with confidence or **Not sure** capture.
5. Practice tray sort with immediate error-specific feedback.
6. Weak-item review for missed or uncertain instruments if time permits.
7. No-hints post-test on a comparable tray variant.
8. Short interview about confusing instruments, feedback usefulness, and perceived realism.

Primary measures are pre/post tray accuracy, duration, missing/wrong/extra/ misidentified/wrong-count errors, confidence calibration, high-confidence errors, low-confidence correct answers, and weak-item recovery. The study supports the training claim if most participants improve accuracy without a severe-error increase, maintain or improve completion time, and can explain at least one corrected error after practice.

The primary value of the novice study is **instrument validation**, not population-effect estimation. The study answers five prerequisite questions that apply to any subsequent population:

1. Do the error categories behave as expected (lookalike confusion vs. missing vs. extra items), or does the scoring rubric produce noise?
2. Does the convergent photo-identification test show a moderate-to-strong correlation with tray-sort accuracy ($r \geq 0.50$)? If not, the tray scores do not measure instrument knowledge in *any* population.
3. Is confidence calibration measurable (high-confidence error rate, **Not sure** frequency), or do participants all guess at the same rate?
4. Does usability friction (scan failures, card confusion, UI navigation) dominate the learning signal, or can participants interact with the system as intended?
5. Which distractors are effective, which feedback messages help, and which instruments are confusing? These content-refinement findings are transferable to an SPD module.

If these fail in novices, they fail everywhere, because they are properties of the measurement instrument, not of the participant population. An SPD pilot run before the measurement tool is validated produces ambiguous results: a null finding could mean the system does not help, or it could mean the scoring rubric, error categories, or convergent measure are inadequate. The novice study decouples these two failure modes.

A corollary is that the novice study does **not** establish the effect size for SPD workers. The dependent variable shifts from absolute accuracy gain (novices) to error-type reduction, confidence calibration, and

retention maintenance (workers) (Section 11.4). The novice results warrant a worker study by showing the instrument functions; they do not substitute for a worker study.

11.4 Study Ladder

Table 28: Study ladder by participant type and validity source

Study	Participants	Validity source	Kirkpatrick level
Study 0: marker reliability	1-2 project members, no learning claim	Measurement reliability (prerequisite)	–
Study 1: novice simulated learning	8-12 college students or classmates with no SPD experience	Response process, internal structure, relations to other variables (convergent)	Level 2 (Learning)
Study 2: delayed retention	Same Study 1 participants, 7 days later	Relations to other variables (temporal)	Level 2 (Learning, durability)
Study 3: expert review	1 SPD educator, instructor, or knowledgeable reviewer	Content	Level 2 (Learning, content plausibility)
Study 4: future SPD pilot	SPD trainees or technicians	Consequential, extrapolation to target context	Level 3 (Behavior)

The project should prioritize Studies 0-2 and, if possible, one expert review.

11.5 Study 0: Marker Reliability

Purpose: prevent app or camera failures from contaminating learner scores. Setup:

- Print all AprilTag cards used in the tray module.
- Use the same device, camera angle, lighting, tray boundary, and table setup planned for the user study.
- Test cards individually, in valid tray layouts, and in cluttered layouts with distractors.

Record total cards visible, cards detected, false positives, missed cards, duplicate or unstable detections, and lighting or occlusion notes. Acceptance target before user testing:

- at least 95% card detection in normal layouts;
- no systematic confusion between marker IDs;
- visible confirmation screen before final scoring so camera misses are not counted as learner errors.

11.6 Study 1: Novice Simulated Learning

Recruit 8-12 college students or classmates with no formal SPD experience and no prior exposure to the module. Collect optional background variables such as medical, biology, lab, tool, or surgical-instrument familiarity. This matches the NeuroVase-style proxy choice: students are not SPD workers, but they are acceptable for a first novice learning and usability study if the claim is bounded to simulated learning. Use a within-subject pre/post design:

intake -> pre-test -> training -> practice -> immediate post-test -> survey/interview

Target one 60 minute session per participant:

Table 29: Study 1: novice session timeline

Segment	Time	Activity
Consent and intake	3-5 min	Explain simulated study, collect background.
Device orientation	2 min	Show how to scan and submit without teaching tray answers.
Pre-test tray sort	5-8 min	No hints, no feedback, timed.
Retrieval cards	10-12 min	Prompt-before-reveal study of required items, distractors, features, and counts.
Quiz	5-8 min	Short recall/recognition quiz with confidence or Not sure .
Practice sort	8-10 min	Same style task with immediate feedback.
Immediate post-test	5-8 min	Parallel variant, no hints, no feedback, timed.
Convergent photo-ID test	5-7 min	Paper-based photo identification of each required instrument plus 2 unseen transfer-probe instruments.
Survey and interview	5-8 min	SUS or short 5-point ease/usefulness/confidence survey plus open questions.

Primary learning measures:

- tray accuracy: required items correctly selected with correct count;
- severe errors: missing, wrong, misidentified, extra, wrong-count;
- duration from task start to final submission;
- high-confidence errors;
- **Not sure** selections or low-confidence correct answers;
- weak-item recovery from quiz/practice to post-test.

Convergent evidence measures:

- **Photo-identification test:** after the post-test tray sort, learners complete a short paper-based test showing high-resolution photos of each required instrument and 2-3 distractor instruments. For each photo, they write the instrument name, indicate whether it belongs in the tray, and rate their confidence. This test shares the knowledge domain (instrument identity, aliases, lookalike discrimination) with the tray sort while differing in response format (identification without physical card selection). The target correlation between post-test tray accuracy and photo-ID accuracy is $r \geq 0.50$, following the convergent-validity standard that two measures of the same construct should show moderate-to-strong association [102, 103, 109].
- **Transfer probe:** the photo-ID test includes 2 instruments from a different tray module that the learner has never seen. If TrayGuard teaches general instrument-discrimination skill rather than module-specific memorization, learners who performed well on the trained tray should show better discrimination on unseen instruments. This is a weak test (novices have no basis to identify unseen

instruments), but the pattern across participants is informative: even chance-level performance on the transfer probe, when combined with strong within-module gains, supports the bounded claim of module-specific learning.

Usability and experience measures:

- completion rate;
- help requests;
- scan failures or rescan attempts;
- SUS or short 5-point ease/usefulness/confidence survey;
- open feedback on confusing cards, confusing instruments, and perceived realism.

This design follows precedents including NeuroVase (tangible AR medical learning with 40 participants, pre/post testing, SUS) [70], Tulipan et al. (Touch Surgery with novice medical students) [78], and Ofstead et al. (SP professional borescope training pilot) [24]. A future between-group design could isolate the incremental effect of simulated tray practice beyond flashcards alone, but the within-subject pre/post design provides simpler logistics and smaller sample requirements for the first prototype.

11.7 Study 2: Delayed Retention

Same-day post-testing supports immediate learning only. It should not be called retention [97]. The delayed check distinguishes immediate practice gain from retained learning. In Kane’s validity argument framework, the extrapolation inference from test score to real-world competence depends on evidence that performance persists after a delay during which transient factors (memorization of exact layouts, test familiarity, short-term motivational effects) have decayed [110]. Without a delayed measure, the interpretive argument supports only a same-day practice-effect claim.

The primary retention interval is **7 days** from the initial session. This interval is a practical compromise: Bell et al. found that medical-knowledge gains from a single intervention could decay measurably within 7 days [97], and Larsen et al. showed that the testing-effect advantage over restudy persisted at least 7 days in a medical-education randomized trial [94]. Ofstead et al. used a longer (2-month) interval with a booster session and reported approximately 90% retention of the post-test gain at that follow-up, but their intervention included workplace homework between sessions [24]. For a single-session training tool without between-session homework, 7 days is the shortest credible retention interval that separates durable learning from ephemeral practice gain.

Procedure:

1. Ask whether the participant studied the material since the first session.
2. Run a no-hints delayed tray sort with a third parallel variant.
3. Capture confidence or **Not sure**.
4. Run the photo-identification test again to measure convergent-retention correlation.
5. Ask what they remembered and what decayed.

Falsification criteria. The retention claim is supported only if:

- delayed accuracy retains at least half of the immediate pre/post gain (i.e., $\text{delayed} - \text{pre} \geq 0.5 \times (\text{post} - \text{pre})$);
- delayed accuracy does not return to within 5 percentage points of the pre-test baseline (i.e., the retention effect is not attributable to measurement noise or test-retest practice);

- at least 70% of Study 1 participants complete Study 2.

If these criteria are not met, the project claim reverts to immediate learning only and the retention gap is reported as a limitation that future work (addressable by distributed practice, multiple training sessions, or a booster) must close. This follows Kane’s principle that the interpretive argument must include explicit rebuttals: the conditions under which the claim would not be supported [110].

Optional extended intervals. If the 7-day check succeeds and schedule permits, additional checks at 21-28 days or 2 months would strengthen the retention claim further. The 2-month interval would match the Ofstead booster precedent [24].

11.8 Study 3: Expert Review

Expert review addresses the main weakness of college-student testing: content validity. The reviewer can be an SPD educator, SPD supervisor, sterile-processing instructor, surgical technology instructor, or faculty member with relevant instrument knowledge. Materials to show:

- tray list;
- instrument cards;
- distractor list;
- aliases and distinguishing features;
- scoring rubric;
- sample learner report.

Questions:

1. Are the instrument names and aliases plausible?
2. Are the required counts and distractors plausible for a training tray?
3. Are the lookalike pairs educationally meaningful?
4. Are any feedback messages misleading or unsafe?
5. Would the weak-item report help target coaching?
6. What would need to change before using this with SPD trainees?

11.9 Analysis and Acceptance Bar

For 8-12 participants, report descriptive paired results:

- participant-level pre/post table;
- mean and median accuracy change;
- mean and median duration change;
- total error counts by type before and after;
- high-confidence errors before and after;
- weak-item examples;
- representative qualitative feedback.

Avoid strong inferential claims unless the sample is larger. If the sample is large enough, add paired differences with confidence intervals. The training claim is supported if most learners show:

- higher post-test accuracy;
- equal or lower post-test duration without accuracy decline;
- fewer severe errors, especially missing and misidentified items;
- fewer high-confidence errors;
- fewer **Not sure** responses or low-confidence correct answers;
- evidence that missed or uncertain items improve after repeated retrieval;
- interview evidence that they can name what they learned or what remains hard.

Best project claim wording:

In a novice college-student sample, TrayGuard improved immediate simulated tray-sorting performance after retrieval-centered practice. A delayed 7-day check tested short-term retention. A convergent photo-identification test provided evidence that the tray-sort scores reflect instrument knowledge rather than task-specific familiarity. Expert review was used to assess whether the tray content and feedback were plausible for future SPD trainee validation. These findings establish that the learning mechanism works in this task domain for a naive population. They do not establish the effect size for experienced SPD workers, whose baseline on familiar trays may be high and for whom the dependent variable shifts from accuracy gain to error-type reduction, confidence calibration, and retention maintenance (Section 11.4).

11.10 Validity Claims Summary

No single study provides all validity sources; the overall claim rests on the accumulated pattern [103, 110].

Table 30: Validity claims mapped to evidence sources

Claim	Validity source	Evidence	Would be falsified by
The tray module content is plausible for training.	Content	Study 3 expert review; instructor-verified module; literature grounding and real count-sheet fields (Section 9.8).	Expert finds names, counts, distractors, or feedback unrealistic; module uses content that contradicts published tray listings.
Learners interact with the system as intended; confidence reflects genuine self-assessment.	Response process	Study 1 usability observation; confidence-calibration analysis (high-confidence error rate, Not sure frequency); SUS scores; interview feedback about confusion points.	Frequent scan failures produce scores that reflect app error rather than learner knowledge; confidence ratings do not distinguish correct from incorrect items; SUS scores below 68.
Error categories behave consistently and decompose learning gains.	Internal structure	Study 1-2 pre/post error-type analysis; lookalike pairs produce proportionally more misidentified errors than non-lookalike items; practice reduces specific error categories rather than shifting errors across types.	Error types do not differ between lookalike and non-lookalike pairs; accuracy gain comes entirely from one error category (e.g., fewer missing items) while other categories remain unchanged.
Tray-sort scores measure instrument knowledge and not only task-specific familiarity.	Relations to other variables	Study 1 convergent photo-ID test; correlation between tray-sort accuracy and photo-ID accuracy; transfer probe on unseen instruments.	Photo-ID correlation $r < 0.30$; photo-ID scores show ceiling or floor effects that prevent discrimination; transfer-probe performance is negatively correlated with trained-tray performance.
Learning persists beyond the initial session.	Relations to other variables (temporal)	Study 2 delayed tray sort at 7 days; retention relative to pre-test baseline; comparison of delayed accuracy to immediate post-test accuracy.	Delayed accuracy returns to within 5 pp of pre-test; less than half of the immediate gain is retained; fewer than 70% of Study 1 participants return.
The report supports instructor coaching without supporting overclaim.	Consequences	Scope boundaries (Section 12); report labels frame results as simulated practice evidence; export fields include weak items and confidence calibration.	Report language uses pass/fail or clinical-readiness labels; results are used for employment decisions; learners report feeling surveilled rather than coached.
Learning gains in novices transfer to SPD workers (extrapolation).	Generalizability / extrapolation	Ofstead precedent: experienced SP workers were well below ceiling on novel tasks; mechanism (retrieval practice + feedback) is population-independent per the training literature; dependent	Worker study fails to show improvement on any worker-relevant outcome; novice-effect-size extrapolation is cited without worker data.

11.11 Future SPD Workplace Pilot

A later SPD pilot requires local count sheets, local photos or real teaching instruments, SPD educator review, technician or trainee participants, approval/privacy review, and a comparison condition such as current study materials or clinical supervisor-led orientation. It should measure whether the system helps target users in a workplace-relevant training context. Only that later pilot can support claims about SPD trainee usefulness, supervised transfer, or operational training adoption.

The SPD pilot must change the dependent variable from the novice study. Where novices are measured on absolute accuracy gain (pre 0 to post 90), experienced workers are measured on: error-type reduction (fewer specific errors on known trays), speed improvement without accuracy loss, confidence calibration (high-confidence error rate), new-module learning rate (time to criterion on an unfamiliar tray), and retention maintenance (preventing decay on low-frequency trays). The pilot protocol must specify which of these applies and the relevant minimum viable improvement before data collection [108, 72].

12 Risk Assessment And Scope Boundaries

(Matthew, Jacob)

12.1 Formal Risk Register

Table 31: Formal risk register with causes, consequences, and mitigations

Risk	Cause	Consequence	Mitigation	Residual Boundary
Student proxy overclaim	Participants are not SPD workers.	Reviewers reject relevance to target users.	State that students test novice simulated learning; add expert review; reserve SPD claims for future pilot.	No SPD worker effectiveness claim.
Same-day retention mistake	Immediate post-test follows practice.	Report overstates durability.	Add committed 7-day delayed check with falsification criteria; otherwise call it immediate learning only.	Retention claim only after delayed test.
Card physical realism	Cards do not reproduce weight, scale, tactile handling, or real visual lookalikes.	Transfer to real instruments remains unknown.	Use cards as selection/count proxy; add local photos or real teaching instruments later.	No clinical handling competency claim.
Marker detection errors	Lighting, occlusion, angle, or overlap causes missed detections.	App errors look like learner errors.	Run Study 0; log raw detections; show confirmation before scoring.	Marker reliability must be reported.
Answer leakage from cards	Labels or marker IDs reveal the answer.	Measured gain is not learning.	Use neutral assessment fronts; randomize marker IDs; hide answer text in assessment.	Study and assessment cards must be separated.
Content invalidity	Student-built module may use wrong names, counts, or distractors.	Results do not reflect plausible SPD training.	Require expert review of module content and feedback.	Content validity limited until SPD review.
Feedback contaminates assessment	Practice answers leak into post-test.	Pre/post comparison inflated.	Use parallel variants; suppress hints and feedback in assessment modes.	Cannot claim broad generalization from one tray.
Usability friction dominates learning	Scanning or workflow confusion slows participants.	Poor scores reflect interface friction.	Track help requests, scan failures, SUS/ease ratings, and qualitative feedback.	Usability results must be reported separately.

12.2 Unknowns and Concerns

The current design is viable as a low-cost formative learning prototype, but several concerns remain unresolved:

Table 32: Unknowns and concerns with strengthening actions

Concern	Why it matters	Strengthening action
Card-to-instrument transfer	AprilTag cards preserve selection and counting decisions, but they do not reproduce weight, scale, handling, tactile inspection, or real visual lookalikes.	Treat the first study as simulated learning only; add an SPD educator review and a later task using local photos or real teaching instruments.
Flashcard overreach	Flashcards support recall, but tray work requires applying local rules under distractor pressure.	Keep flashcards as preparation, then make the main outcome a no-hints tray sort with missing, extra, wrong, misidentified, and wrong-count scoring.
Marker identity leakage	If cards visibly encode names or IDs, learners may match labels instead of learning instruments.	Use neutral card fronts during assessment, randomize marker IDs, hide answer text, and separate study cards from assessment cards.
Content validity	A student-built tray module may contain wrong names, unrealistic distractors, or nonlocal quantities.	Require instructor or SPD educator verification of every instrument, alias, count, and distractor before using results as evidence.
Assessment equivalence	Pre/post gains are weak evidence if the post-test is easier or repeats the same exact layout.	Build parallel pre/post tray variants with matched item families, counts, distractors, and difficulty.
Short-term learning only	A same-day post-test can show practice effects rather than retention.	Add a delayed retention check or booster if schedule permits.
Novice participant limits	Student participants can test learnability, but they do not represent SPD technicians under workflow pressure.	Report student results as novice simulated-learning evidence and reserve adoption/clinical-transfer claims for an SPD pilot.
Scanning reliability	Camera angle, lighting, occlusion, and card overlap may produce app errors that look like learner errors.	Log raw detections, show a confirmation state before scoring, and run a marker-detection bench test before the user study.
Confidence calibration	Confidence ratings can become noise if captured too often or too vaguely; novice learners may also exhibit systematic overconfidence on items they cannot reliably identify [111].	Capture simple confidence or Not sure at assessment moments and analyze high-confidence errors separately.
Instructor usefulness	Better learner scores do not automatically mean the output helps educators.	Add a short instructor-facing report and ask an educator whether the weak-item summary would change coaching.

These holes do not invalidate the design. They define the claim boundary: the prototype can show whether a marker-card training loop improves novice performance on a simulated local tray task. It cannot yet show real SPD competence, reduced OR delays, or deployment-ready instrument recognition.

12.3 Scope Boundaries

Allowed after Study 1:

- novices improved on a simulated tray task;
- the interface was usable for first-time learners;
- retrieval cards plus practice sorting produced measurable immediate gains.

Allowed after Study 2:

- gains persisted after 7 days (meeting the falsification criteria in Section 11.7);
- the system showed measurable short-term retention in a simulated tray task, distinguishable from same-day practice effects.

Allowed after Study 3:

- an expert reviewer found the module plausible for future SPD validation;
- content concerns were identified and revised before broader testing.

Not allowed yet:

- SPD technicians will improve;
- the system reduces tray errors in hospitals;
- the system reduces OR delays;
- the system certifies competency;
- the app recognizes real surgical instruments;
- same-day gains demonstrate retention.

13 Broader Impacts And Ethics

(Matthew, Jacob)

TrayGuard’s positive impact is a more accessible way to practice local tray knowledge before a learner consumes scarce supervised training time. If it works, learners get repeatable practice, instructors get clearer weak-item evidence, and future teams get a safer bridge between training content and camera-supported tray review. The ethical risk is that a useful training report could be misused. A learner score is not a certification result, employment screen, disciplinary record, or permission to perform unsupervised sterile-processing work. The report should therefore label outputs as simulated practice evidence and should avoid language such as pass, fail, competent, or cleared unless a formal program adds its own supervised competency process. Using the principles of beneficence, nonmaleficence, autonomy, and justice:

Table 33: Bioethics principles applied to TrayGuard

Principle	TrayGuard implication
Beneficence	The system should help learners improve and help educators target coaching, especially on missing, wrong, misidentified, and high-confidence errors.
Nonmaleficence	The system should not imply clinical readiness, replace hands-on supervision, or hide uncertainty behind a single score.
Autonomy	Learners and instructors should understand what is being recorded, why it is being recorded, and how the report will be used.
Justice	Access should not depend on having a perfect local instrument library, expensive hardware, or one preferred training site. Modules should be updateable and printable.

Negative externalities and controls:

Table 34: Negative externalities and ethical controls

Risk	Ethical concern	Control
Punitive learner metrics	Scores could become surveillance or discipline instead of coaching.	Use hashed IDs for studies; report weak items and practice progress; prohibit employment or competency decisions from prototype data.
Overreliance on simulated scores	A high score could be mistaken for real SPD readiness.	Label all results as simulated; require supervised hands-on validation for clinical claims.
Reduced supervised practice	Programs might use the tool to cut clinical supervisor time.	Frame TrayGuard as preparation for supervised practice, not replacement.
Unequal local content	Better-resourced sites may build richer modules.	Use open file formats, printable cards, and a minimal module template.
Instrument-library bias	Learners may only practice instruments represented in the module.	Version modules, list coverage gaps, and require local review before use.
Answer leakage	Labels or marker IDs could inflate learning results.	Use neutral assessment cards and hide names/counts in assessment modes.
Printing and material waste	Card iterations create paper/plastic waste.	Use reusable sleeves, print small modules, and revise digitally before reprinting.
Privacy creep	Future reports could collect names, staff IDs, or workplace behavior.	Keep patient data out of scope; collect only fields needed for learning analysis.

These controls are part of the design, not optional documentation. They protect the stakeholder value of TrayGuard by keeping the training tool honest about what it can and cannot prove.

14 Project Plan And Future Work

(Matthew, Jacob)

14.1 Completed Work

The following subsystems have been implemented:

- **Training web app.** A FastAPI backend serves a single-page browser application with seven-step learner flow: intake, pre-test, study cards, quiz, practice sort, post-test, and summary. Camera capture through `getUserMedia` streams frames to a server-side ArUco detection endpoint. Manual quantity entry serves as a fallback when camera access is unavailable or unreliable `src/trayguard/web/app.py`.
- **Tray module data model.** Eight file-backed tray modules define required instruments, distractors, lookalike pairs, assessment variants, and ArUco marker mappings. The primary module, `basic_general_tray_v1`, contains 10 required instrument types (15 total required units), 5 distractors, 6 lookalike pairs, and 2 assessment variants with matched difficulty. Five instrument catalogs covering 400+ classes provide authoring reference material `config/tray_modules/`.
- **Card generation and printing.** A CLI command generates printable ArUco marker PNGs, a marker-grid index sheet for camera scanning, and double-sided study flashcards with instrument images, names, aliases, distinguishing features, and tray-specific quantities. An image-processing pipeline handles EXIF rotation, content-aware cropping, and automatic marker placement `src/trayguard/learning/cards.py`.
- **Scoring engine.** A scoring function classifies each item into correct, missing, extra, wrong, misidentified (lookalike substitution), or wrong-count categories. Error-point aggregation maps to an accuracy score and required-recall metric. Confidence ratings (1-5 scale and a `Not sure` checkbox) are captured per item and flagged as high-confidence errors or low-confidence correct answers `src/trayguard/learning/scoring.py`.
- **Export and run storage.** Each completed run produces a JSON payload plus CSV files at the attempt level and item level. Learner identifiers are hashed (SHA-256 truncated to 16 hex characters). The export schema includes all error categories, confidence metadata, and module version fields `src/trayguard/learning/export.py`; `src/trayguard/learning/run_store.py`.
- **Tests.** Unit tests verify module loading, scoring accuracy (full-credit and lookalike-substitution cases), CSV/JSON export file completeness, and ArUco marker generation and detection `tests/test_learning.py`.

14.2 Project Plan Postmortem

The training-centered pivot (Section 6) reduced risk by changing the main proof obligation from deployment-grade recognition to measurable novice learning.

14.3 Future Work Roadmap

Table 35: Future work roadmap phases

Phase	Goal	Exit Criteria
1. Minimal prototype	Build the end-to-end local tray loop.	Load module, scan cards, run pre-test, show study cards, run quiz, run practice sort, run post-test, export metrics.
2. Verified module	Make the first tray content defensible.	8-12 required instruments, distractors, lookalikes, neutral assessment cards, matched variants, and expert/instructor review.
3. Marker reliability	Prove the app can observe card choices.	At least 95% detection in normal layouts, no systematic ID confusion, confirmation screen before scoring.
4. Novice study	Test immediate simulated learning.	8-12 college-student participants, pre/post results, usability notes, and bounded claim language.
5. Retention check	Test whether learning persists.	7-day delayed no-hints tray sort with third matched variant, plus falsification criteria (Section 11.7).
6. Report refinement	Turn study output into project evidence.	Participant-level results, aggregate pre/post plots, error-category table, high-confidence errors, and qualitative themes.
7. SPD educator review	Improve content validity.	Reviewer feedback on names, counts, distractors, lookalikes, scoring, and weak-item report.
8. Future SPD pilot	Test target-user relevance.	SPD trainee or technician participants, local count sheet/photos, approval/privacy review, and comparison condition.
9. Optional CV support	Reintroduce real-instrument recognition as support, not the core proof.	Detector tested on local photos with generalization and failure-mode reporting.

14.4 GitLab Epic / Issue Map

Table 36: Epic and issue map for continuation

Epic	Issues
Training Module	Define first tray module; add instrument cards with aliases and features; add tray quantities and distractors; build study-card view; build identification quiz.
Scoring And Feedback	Record quiz correctness, duration, and confidence; build simulated tray sorting; implement missing/extra/wrong/misidentified/wrong-count scoring; add practice feedback.
Assessment	Build pre-test mode; build post-test mode; suppress hints and feedback during assessment; add delayed-retention mode if schedule allows.
Evidence And Reporting	Export learner metrics; generate pre/post summary; report confidence calibration, high-confidence errors, and weak-item recovery.
Pilot Evidence	Write pilot walkthrough script; run marker reliability check; run novice study; record confusion points; add limitation note about non-SPD participants.
Content Validation	Prepare expert-review packet; collect reviewer feedback; revise names, counts, distractors, and feedback language.
Presentation	Lock final demo narrative around simulated learning, retention boundary, and future SPD validation.

14.5 Next Team Priorities

14.6 Budget

Table 37: Project budget by category

Category	Expected Cost	Notes
Software prototype	\$0 direct cost	Uses existing local app stack and open-source AprilTag support where available.
Physical cards	\$5-25	Printer paper/cardstock, sleeves, tape, or foam backing.
Tray surface	\$0-30	Existing table, poster board, foam board, or simple marked tray boundary.
Camera/device	\$0 if using existing laptop/tablet/phone	Optional stand improves reliability but is not required for the first study.
Participant cost	\$0-100	Small snacks/gift cards if allowed; class demo can be unpaid.
Expert review	\$0-200	Likely unpaid instructor feedback; paid SPD educator review would be stronger.
Future SPD pilot	TBD	Depends on site approval, staff time, privacy review, and local instrument access.

The main cost is labor: module authoring, prototype implementation, marker testing, study moderation, analysis, and report writing.

15 Discussion And Final Recommendations

(Matthew, Jacob)

This report supports a clear next step: build and test the software-first training loop before investing more effort in deployment-grade tray automation. The problem evidence is strong enough to justify the target task, and the learning evidence is strong enough to justify a simulated training prototype. The current evidence is not strong enough to claim that TrayGuard reduces hospital tray errors, improves SPD technician performance, or recognizes real surgical instruments in clinical conditions.

The evaluation plan targets Kirkpatrick Level 2 (Learning) with internal- structure, convergent, and temporal evidence; Level 3 and Level 4 remain future work [108]. The five-source validity framework provides the evidentiary structure with explicit falsification criteria [103].

Population generalizability is addressed by separating the dependent variable by population: novices on absolute accuracy gain, workers on error-type reduction, confidence calibration, and new-module learning rate (Section 11.4). The Ofstead precedent confirms that even certified SP workers have room for improvement on novel tasks [24].

Stakeholders can conclude the following:

- Tray reconstruction is a defensible focus because many readiness failures involve assembly, visualization, identification, missing items, wrong items, extra items, and local count-sheet use.

- The training pivot is justified because local tray knowledge and supervised practice are real bottlenecks, while deployment-grade CV has a much larger validation burden.
- A marker-card prototype can support a bounded novice-learning study if the cards are neutral, detection is reliable, assessment variants are comparable, and results are labeled as simulated.
- The project has reduced concept risk, requirement risk, scoring risk, and study design risk by defining the module contract, performance targets, error categories, study ladder, convergent validity target, and claim boundaries.
- The evaluation design addresses content validity (expert review), internal structure (error-type analysis), relations to other variables (convergent photo-ID test), and temporal persistence (committed 7-day retention check with falsification criteria).

Stakeholders cannot yet conclude that TrayGuard:

- certifies sterile-processing competence;
- replaces hands-on hours, clinical supervisor observation, or local sign-off;
- reduces OR delays or hospital tray-defect rates;
- works for SPD technicians in a real workplace;
- recognizes real instruments across manufacturers, lighting, occlusion, and tray layouts;
- produces the same effect size for experienced workers as for novices (the dependent variable and expected magnitude differ by population);
- supports retention unless the 7-day delayed check meets the falsification criteria specified in Section 11.7.

The next team should therefore implement the learning loop described in Section 7, extended with a convergent photo-ID test. After that loop works, the team should run the marker reliability check, then a small novice pre/post study with the convergent photo-ID measure, then the committed 7-day retention check, then an expert review of names, counts, distractors, lookalikes, and feedback. Only after those results should the project expand toward real teaching instruments, local photos, SPD trainees, or real-instrument CV. Final recommendation:

Treat TrayGuard as a local tray learning and assessment platform first. Use the project prototype to prove that novices can improve on a simulated local tray task, demonstrate that the gains reflect instrument knowledge (convergent photo-ID evidence), persist for at least 7 days (retention check), and are grounded in plausible content (expert review), then use those results to decide whether a future SPD pilot or camera-supported tray review system is worth the additional validation cost.

16 References

References

- [1] Alfred et al., “Work systems analysis of sterile processing: assembly,” *BMJ Quality & Safety*, 2021. [Online]. Available: <https://qualitysafety.bmj.com/content/30/4/271>
- [2] Zhu et al., “Errors in packaging surgical instruments based on a surgical instrument tracking system: an observational study,” *BMC Health Services Research*, 2019. [Online]. Available: <https://link.springer.com/article/10.1016/j.arth.2022.08.017>

- [3] Nichol et al., "Observed rates of surgical instrument errors point to visualization tasks as being a critically vulnerable point in sterile processing and a significant cause of lost chargeable or minutes," *BMC Surgery*, 2024. [Online]. Available: <https://link.springer.com/article/10.1186/s12893-024-02407-1>
- [4] Nichol and Saari, "Patterns in staff reported surgical instrument errors point to failures in visualization as a critically weak point in sterile processing of surgical instruments," *Perioperative Care and Operating Room Management*, 2023. [Online]. Available: <https://doi.org/10.1016/j.pcorn.2023.100356>
- [5] George et al., "A day in the life of a surgical instrument: The cycle of sterilization," *Annals of Surgery Open*, 2024. [Online]. Available: <https://doi.org/10.1097/AS9.0000000000000381>
- [6] Nichol and Saari, "Risk modeling of errors in the surgical instrument cycle, insights into solutions for an expensive and persistent problem," *Perioperative Care and Operating Room Management*, 2023. [Online]. Available: <https://doi.org/10.1016/j.pcorn.2023.100333>
- [7] Pennsylvania Patient Safety Authority, "Contaminated surgical instruments: The worst-case scenario," *Pennsylvania Patient Safety Advisory*, 2006. [Online]. Available: https://patientsafety.pa.gov/ADVISORIES/Pages/200603_20.aspx
- [8] Association of Health Care Journalists, "Hospital inspection report detail snxu11," 2024, accessed May 28, 2026. [Online]. Available: <https://www.hospitalinspections.org/report-detail/SNXU11>
- [9] Ingold, "Colorado's largest hospital had to pause surgeries for a week following a complaint. here's what health inspectors saw." *The Colorado Sun*, 2025. [Online]. Available: <https://coloradosun.com/2025/10/16/uchealth-university-of-colorado-hospital-surgery-pause/>
- [10] Abbott et al., "The lifetime risk of surgery in england: a nationwide observational cohort study," *British Journal of Anaesthesia*, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11443128/>
- [11] Lind, "Instrument error! what happened? identifying causes, adopting solutions," *PROCESS: The Magazine for Sterile Processing*, 2026. [Online]. Available: <https://elearning.myhspa.org/cis-lesson-plans>
- [12] N. Chobin, "Instrument-marking methods must be maintained properly," *Infection Control Today*, 2017, accessed May 28, 2026. [Online]. Available: <https://www.infectioncontrolday.com/view/instrument-marking-methods-must-be-maintained-properly>
- [13] HSPA Surgical Instrument Resources, "Surgical instrument resources," 2026, accessed May 28, 2026. [Online]. Available: <https://myhspa.org/education/publications/surgical-instrument-resources/>
- [14] American Society for Healthcare Central Service Professionals (ASHCSP), "Central service department history and development," 2007, describes the historical evolution of central service from decentralized processing to a specialized department.
- [15] The Joint Commission, "Sentinel event data 2024 annual review," 2025, accessed May 28, 2026. [Online]. Available: <https://www.jointcommission.org>
- [16] dos Santos et al., "Approaches to the rationalization of surgical instrument trays: scoping review and research agenda," *BMC Health Services Research*, 2021. [Online]. Available: <https://bmchealthservres.biomedcentral.com/articles/10.1186/s12913-021-06142-8>
- [17] Ahmadi et al., "A data-driven decision-making model for configuring surgical trays based on the likelihood of instrument usages," *Mathematics*, 2023. [Online]. Available: <https://www.mdpi.com/2227-7390/11/9/2219>
- [18] CDC, "Sterilizing practices," *Guideline for Disinfection and Sterilization in Healthcare Facilities*, 2008. [Online]. Available: <https://www.cdc.gov/infection-control/hcp/disinfection-sterilization/sterilizing-practices.html>
- [19] Castillo-Gutierrez, "Sharpening sensory skills in sp technicians to support quality and safety," *PROCESS: The Magazine for Sterile Processing*, 2025. [Online]. Available: <https://www.eventreg.purdue.edu/info/sterile-processing/pdf/crcst-201.pdf>
- [20] Fast et al., "The impact of a short-term training program on workers' sterile processing knowledge and practices in 12 ethiopian hospitals: A mixed methods study," *PLOS ONE*, 2019. [Online]. Available: <https://doi.org/10.1371/journal.pone.0215643>
- [21] Shreckengost et al., "Clean and confident: Impact of sterile instrument processing workshops on knowledge and confidence in five low- and middle-income countries," *Surgical Infections*, 2022. [Online]. Available: <https://doi.org/10.1089/sur.2021.187>

- [22] HSPA CRCST, "Certified registered central service technician (crcst)," 2026, accessed May 28, 2026. [Online]. Available: <https://myhspa.org/certification/certification-overview/certified-registered-central-service-technician-crcst/>
- [23] Nadeau, "Back to basics in the spd," *Healthcare Purchasing News*, 2024. [Online]. Available: <https://www.hpnonline.com/sterile-processing/article/55247421/back-to-basics-in-the-spd>
- [24] Ofstead et al., "Improving mastery and retention of knowledge and complex skills among sterile processing professionals: A pilot study on borescope training and competency testing," *American Journal of Infection Control*, 2023. [Online]. Available: <https://doi.org/10.1016/j.ajic.2023.03.002>
- [25] AORN Staff, "5 proven ways to address your sterile processing staffing shortage," 2025, accessed May 28, 2026. [Online]. Available: <https://www.aorn.org/about-aorn/aorn-newsroom/periop-life/article/5-proven-ways-to-address-your-sterile-processing-staffing-shortage>
- [26] Kovach, "Sterile processing certification, does it have value?" *Infection Control Today*, 2012. [Online]. Available: <https://www.infectioncontrolday.com/view/sterile-processing-certification-does-it-have-value>
- [27] Nadeau, "Your cs/spd staff is certifiable," *Healthcare Purchasing News*, 2017. [Online]. Available: <https://www.hpnonline.com/sterile-processing/article/13000740/your-cs-spd-staff-is-certifiable>
- [28] Hu et al., "Improvement and implementation of central sterile supply department training program based on action research," *BMC Nursing*, 2024. [Online]. Available: <https://link.springer.com/article/10.1186/s12912-024-01809-z>
- [29] Thurmond, "Power of competency checklists for sterile processing," *Healthcare Purchasing News*, 2020. [Online]. Available: <https://www.hpnonline.com/sterile-processing/article/21202328/power-of-competency-checklists-for-sterile-processing>
- [30] Chen et al., "Incidence of adverse events in central sterile supply department: A single-center retrospective study," *Risk Management and Healthcare Policy*, 2023. [Online]. Available: <https://doi.org/10.2147/RMHP.S423108>
- [31] J. Kusler-Jensen, "Sterile processing department benchmarking for labor productivity," *AORN Journal*, vol. 118, no. 2, pp. 94–100, 2023, describes the AAMI benchmarking framework's worked-hours-per-unit (WHPU) as the primary productivity metric for SPD.
- [32] Lind, "Strategies to enhance the surgical instrument system," *PROCESS: The Magazine for Sterile Processing*, 2025. [Online]. Available: https://www.eventreg.purdue.edu/info/sterile-processing/pdf/cis/SeptOct2025_CIS.pdf
- [33] STERIS, "How to track loaner surgical instrument trays," 2021, accessed May 28, 2026. [Online]. Available: <https://www.steris.com/healthcare/knowledge-center/sterile-processing/how-to-track-loaner-surgical-instrument-trays>
- [34] Huang et al., "Situations and demands of central sterile supply department training on nursing interruptions," *BMC Health Services Research*, 2025. [Online]. Available: <https://link.springer.com/article/10.1186/s12913-024-12190-7>
- [35] OSHA Central Sterile Supply, "Hospitals etool: Central sterile supply - work-related musculoskeletal disorders," 2026, accessed May 28, 2026. [Online]. Available: <https://www.osha.gov/etools/hospitals/central-supply/work-related-musculoskeletal-disorders>
- [36] B. McClenathan, "Unseen but essential: The sterile processing knowledge demand and staffing crisis," 2025, accessed May 28, 2026. [Online]. Available: <https://www.surgicaldirections.com/insights/whitepaper/2025/09/09/unseen-but-essential-the-sterile-processing-knowledge-demand-and-staffing-crisis/>
- [37] STERIS, "Capital budgeting in sterile processing," 2024, accessed May 28, 2026. [Online]. Available: <https://www.steris.com/healthcare/knowledge-center/sterile-processing/capital-budgeting-in-sterile-processing>
- [38] Moab Healthcare, "The hunt for revenue and cost reductions," 2024, accessed May 28, 2026. [Online]. Available: <https://moabhealthcare.com/the-hunt-for-revenue-and-cost-reductions/>
- [39] K. Nadeau, "The ups and downs of spd from 2022 to 2023," *Healthcare Purchasing News*, 2023, accessed May 28, 2026. [Online]. Available: <https://www.hpnonline.com/sterile-processing/article/53060535/the-ups-and-downs-of-spd-from-2022-to-2023>

- [40] M. Perry, "Change what you see in spd," *The Circulating Life*, 2022, accessed May 28, 2026. [Online]. Available: <https://thecirculatinglife.com/2022/10/10/change-what-you-see-in-spd/>
- [41] Mácola et al., "An analysis of the economic challenges facing central sterile processing employees in the united states: Results of a national survey," *Perioperative Care and Operating Room Management*, 2025. [Online]. Available: <https://doi.org/10.1016/j.pcorn.2025.100520>
- [42] Brozak, "Spd staffing issues: 8 tips for recruiting/retaining staff," *Medline*, 2025. [Online]. Available: <https://www.medline.com/strategies/perioperative/spd-staffing-issues-8-tips-for-recruiting-retaining-staff/>
- [43] Evolved Sterile Processing, "Sterile processing staffing woes — what is driving the divide," 2025, accessed May 28, 2026. [Online]. Available: <https://www.evolvedsterileprocessing.com/post/sterile-processing-staffing-woes>
- [44] C. Young, "Travel sterile processing jobs: Why spd talent is leaving — and what hospitals can do about it," 2026, accessed May 28, 2026. [Online]. Available: <https://specialtycareus.com/blog-travel-sterile-processing-jobs/>
- [45] Bureau of Labor Statistics, U.S. Department of Labor, "Union members — 2023," 2024, accessed May 28, 2026. [Online]. Available: https://www.bls.gov/news.release/archives/union2_01232024.pdf
- [46] Loria, "Instrument tray management made easy," *Outpatient Surgery Magazine*, 2024. [Online]. Available: <https://www.aorn.org/outpatient-surgery/article/instrument-tray-management-made-easy>
- [47] Natarus et al., "Optimization of a sterile processing department using lean six sigma methodology, staffing enhancement, and capital investment," *The Joint Commission Journal on Quality and Patient Safety*, 2025. [Online]. Available: <https://doi.org/10.1016/j.jcjq.2024.10.006>
- [48] American Association for Respiratory Care (AARC), "Aarc timeline," 2023, accessed June 2, 2026. [Online]. Available: <https://www.aarc.org/about-us/aarc-timeline/>
- [49] Association of Surgical Technologists (AST), "Surgical technologist and surgical assistant state law map," 2026, accessed June 2, 2026. [Online]. Available: https://www.ast.org/public_policy/map_of_state_laws/
- [50] Amitai Etzioni, *The Semi-Professions and Their Organization: Teachers, Nurses, Social Workers*. New York: Free Press, 1969.
- [51] CBSPD Technician Exam, "Spd technician certification exam c.s.p.d.t." 2026, accessed May 28, 2026. [Online]. Available: <https://www.cbspd.net/technician.htm>
- [52] Towson University, "Sterile processing technician course," 2026, accessed May 28, 2026. [Online]. Available: <https://www.towson.edu/campus/business-public-engagement/continuing-education/health-medical-courses/sterile-processing-technician.html>
- [53] Altamont Healthcare, "Spd: Sterile processing and distribution technician," 2026, accessed May 28, 2026. [Online]. Available: <https://altamonthhealthcare.com/programs/sterile-processing-technician/>
- [54] Klacik, "Core elements of sterile processing training programs," *PROCESS: The Magazine for Sterile Processing*, 2023. [Online]. Available: <https://www.eventreg.purdue.edu/info/central-service/pdf/CIS299.pdf>
- [55] Penn Foster, "Sterile processing technician career diploma," 2026, accessed May 28, 2026. [Online]. Available: <https://www.pennfoster.edu/programs/healthcare/sterile-processing-career-diploma>
- [56] MedCerts Catalog, "Academic catalog 2025," 2025, accessed May 28, 2026. [Online]. Available: <https://medcerts.com/wp-content/uploads/2025/03/MedCerts-Academic-Catalog-2025-01.pdf>
- [57] AIMS Education, "Sterile processing technician training," 2026, accessed May 28, 2026. [Online]. Available: <https://aimseducation.edu/sterile-processing-technician>
- [58] Central Sterilization Solutions, "In-person sterile processing class," 2026, accessed May 28, 2026. [Online]. Available: <https://centralsterilizationschool.com/in-person-crct-class/>
- [59] McBride et al., "Allied health pre-entry student clinical placement capacity: can it be sustained?" *Australian Health Review*, 2020. [Online]. Available: <https://doi.org/10.1071/AH18088>
- [60] North Greater Sacramento COE, "Pathways forward: Collaborative solutions to allied health clinical placement challenges," 2025, accessed May 28, 2026. [Online]. Available: https://coecc.net/wp-content/uploads/2025/04/Pathways-Forward_Report_24-25-1.pdf

- [61] Atabuzzaman et al., “Real-time ultra-fine-grained surgical instrument classification,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025W/FGVC/html/Atabuzzaman_Real-Time_Ultra-Fine-Grained_Surgical_Instrument_Classification_CVPRW_2025_paper.html
- [62] Stryker Gamma4 Count Sheet, “Gamma4 indication tray count sheet,” 2023, accessed May 28, 2026. [Online]. Available: <https://www.stryker.com/content/dam/stryker/australia-new-zealand/trauma/products/gamma4/Gamma4%20Count%20Sheet%20-%20SSP.pdf>
- [63] Stryker IMN Count Sheet, “Imn basic instruments tray count sheet,” 2023, accessed May 28, 2026. [Online]. Available: <https://www.stryker.com/content/dam/stryker/australia-new-zealand/trauma/products/gamma4/IMN%20Basic%20Inst.%20Count%20Sheet%20%28CA4114%29.pdf>
- [64] Rubak et al., “Surgical instrument tray optimization process at a university hospital: A comprehensive overview,” *Surgery Open Science*, 2024. [Online]. Available: <https://doi.org/10.1016/j.sopen.2024.09.007>
- [65] Eussen et al., “Surgical tray optimization: a prospective and survey-based evaluation of environmental and economic outcomes,” *Surgical Endoscopy*, 2026. [Online]. Available: <https://doi.org/10.1007/s00464-025-12499-2>
- [66] Cook et al., “Technology-enhanced simulation for health professions education: a systematic review and meta-analysis,” *JAMA*, 2011. [Online]. Available: <https://jamanetwork.com/journals/jama/fullarticle/1104300>
- [67] McGaghie et al., “Does simulation-based medical education with deliberate practice yield better results than traditional clinical education? a meta-analytic comparative review of the evidence,” *Academic Medicine*, 2011. [Online]. Available: <https://academic.oup.com/academicmedicine/article-abstract/86/6/706/8352665>
- [68] Issenberg et al., “Features and uses of high-fidelity medical simulations that lead to effective learning: a beme systematic review,” *Medical Teacher*, 2005. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/16147767/>
- [69] Barrison et al., “Electronic flashcards in health professions education: A scoping review,” *Academic Medicine*, 2025. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/39774058/>
- [70] Jahani et al., “Neurovase: A tangible mobile augmented reality learning system for neurovascular anatomy and stroke education,” *arXiv preprint*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.00296>
- [71] Karia et al., “The use of qr codes in healthcare education: a scoping review,” *BMC Medical Education*, 2019. [Online]. Available: <https://bmcmmededuc.biomedcentral.com/articles/10.1186/s12909-019-1876-4>
- [72] Arthur et al., “Effectiveness of training in organizations: a meta-analysis of design and evaluation features,” *Journal of Applied Psychology*, 2003. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/12731707/>
- [73] Salas et al., “The science of training and development in organizations: What matters in practice,” *Psychological Science in the Public Interest*, 2012. [Online]. Available: <https://www.psychologicalscience.org/publications/journals/pspi/training-and-development.html>
- [74] Ward et al., “The co-design, implementation and evaluation of a serious board game ‘playdecide patient safety’ to educate junior doctors about patient safety and the importance of reporting safety concerns,” *BMC Medical Education*, 2019. [Online]. Available: <https://bmcmmededuc.biomedcentral.com/articles/10.1186/s12909-019-1655-2>
- [75] Tray Pacer, “Tray pacer system sterile processing surgical tray assembly software,” 2026, accessed May 28, 2026. [Online]. Available: <https://traypacersystem.com/>
- [76] LayerJot SID, “Sid 4.0,” 2026, accessed May 28, 2026. [Online]. Available: <https://www.layerjot.com/sid>
- [77] CensiTrac, “Surgical instrument tracking software | censitrac,” 2026, accessed May 28, 2026. [Online]. Available: <https://censis.com/solutions/censitrac/>
- [78] Tulipan et al., “Touch surgery: Analysis and assessment of validity of a hand surgery simulation ‘app’,” *Hand*, 2019. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/29363359/>
- [79] Medtronic Touch Surgery, “Touch surgery simulations,” 2026, accessed May 28, 2026. [Online]. Available: <https://www.medtronic.com/en-us/healthcare-professionals/specialties/touch-surgery/simulations.html>

- [80] Deol et al., "Artificial intelligence model for automated surgical instrument detection and counting: an experimental proof-of-concept study," *Patient Safety in Surgery*, 2024. [Online]. Available: <https://doi.org/10.1186/s13037-024-00406-y>
- [81] Kienle et al., "Surgical instrument detection on the instrument stand using neural networks," *Current Directions in Biomedical Engineering*, 2025. [Online]. Available: <https://doi.org/10.1515/cdbme-2025-0235>
- [82] da Silva et al., "Towards autonomous instrument tray assembly for sterile processing applications," *arXiv*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.01679>
- [83] Orland et al., "Does virtual reality improve procedural completion and accuracy in an intramedullary tibial nail procedure? a randomized control trial," *Clinical Orthopaedics and Related Research*, 2020. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7431248/>
- [84] Wolters Kluwer Body Interact, "Body interact virtual patient care simulator," 2026, accessed May 28, 2026. [Online]. Available: <https://www.wolterskluwer.com/en/solutions/lippincott-medicine/medical-education/body-interact-virtual-patient-care-simulator>
- [85] Surgical Science, "Simulators," 2026, accessed May 28, 2026. [Online]. Available: <https://surgiscience.com/simulators/>
- [86] Roediger and Karpicke, "Test-enhanced learning: Taking memory tests improves long-term retention," *Psychological Science*, 2006. [Online]. Available: <https://journals.sagepub.com/doi/10.1111/j.1467-9280.2006.01693.x>
- [87] Goddard et al., "Automation bias: a systematic review of frequency, effect mediators, and mitigators," *Journal of the American Medical Informatics Association*, 2012. [Online]. Available: <https://academic.oup.com/jamia/article/19/1/121/732254>
- [88] Kelly, "Human in the loop: Balancing artificial intelligence, clinical judgment, and legal responsibility," *International Journal of Law and Psychiatry*, 2026. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/41932077/>
- [89] Olson, "Apriltag: A robust and flexible visual fiducial system," *2011 IEEE International Conference on Robotics and Automation*, 2011. [Online]. Available: <https://april.eecs.umich.edu/papers/details.php?name=olson2011tags>
- [90] Kalaitzakis et al., "Fiducial markers for pose estimation: Overview, applications and experimental comparison of the artag, apriltag, aruco and stag markers," *Journal of Intelligent and Robotic Systems*, 2021. [Online]. Available: <https://doi.org/10.1007/s10846-020-01307-9>
- [91] Wang et al., "Apriltag 2: Efficient and robust fiducial detection," *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016. [Online]. Available: <https://ieeexplore.ieee.org/document/7759617>
- [92] Krogus et al., "Apriltag 3: A robust, flexible, and faster fiducial marker system," *Technical Report*, University of Michigan, 2019. [Online]. Available: <https://april.eecs.umich.edu/papers/details.php?name=krogus2019april3>
- [93] van Gaalen et al., "Gamification of health professions education: a systematic review," *Advances in Health Sciences Education*, 2021. [Online]. Available: <https://link.springer.com/article/10.1007/s10459-020-10000-3>
- [94] Larsen et al., "Repeated testing improves long-term retention relative to repeated study: A randomised controlled trial," *Medical Education*, 2009. [Online]. Available: <https://doi.org/10.1111/j.1365-2923.2009.03518.x>
- [95] Dunlosky et al., "Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology," *Psychological Science in the Public Interest*, 2013. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/26173288/>
- [96] Hattie and Timperley, "The power of feedback," *Review of Educational Research*, 2007. [Online]. Available: <https://journals.sagepub.com/doi/10.3102/003465430298487>
- [97] Bell et al., "Knowledge retention after an online tutorial: A randomized educational experiment among resident physicians," *Journal of General Internal Medicine*, 2008. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC2517967/>

- [98] SteriBoost, “Vr simulation training,” 2026, accessed May 28, 2026. [Online]. Available: <https://steriboost.com/vr-simulation-training/>
- [99] Vanhaverbeke et al., “The introduction of steridefi: a serious game for continuous education in sterilization,” *Pharmaceutical Technology in Hospital Pharmacy*, 2018. [Online]. Available: <https://www.degruyterbrill.com/document/doi/10.1515/ptph-2018-0009/html>
- [100] Malterud et al., “Sample size in qualitative interview studies: Guided by information power,” *Qualitative Health Research*, 2016. [Online]. Available: <https://doi.org/10.1177/1049732315617444>
- [101] Hennink and Kaiser, “Sample sizes for saturation in qualitative research: A systematic review of empirical tests,” *Social Science & Medicine*, 2022. [Online]. Available: <https://doi.org/10.1016/j.socscimed.2021.114523>
- [102] S. Messick, “Validity,” *Educational Measurement*, pp. 13–103, 1989, unified validity theory: six aspects of construct validity — content, substantive, structural, generalizability, external, and consequential.
- [103] D.A. Cook and T.J. Beckman, “Current concepts in validity and validation for surgeons and health care professionals,” *Journal of the American College of Surgeons*, vol. 204, no. 4, pp. 678–685, 2006, five-source validity framework for simulation-based assessment: content, response process, internal structure, relations to other variables, and consequences.
- [104] Gerhard Pahl, Wolfgang Beitz, Jörg Feldhusen, and Karl-Heinrich Grote, *Engineering Design: A Systematic Approach*, 3rd ed. Springer, 2007.
- [105] Karl T. Ulrich, Steven D. Eppinger, and Maria C. Yang, *Product Design and Development*, 7th ed. McGraw-Hill, 2020.
- [106] Martinengo et al., “Spaced digital education for health professionals: Systematic review and meta-analysis,” *Journal of Medical Internet Research*, 2024. [Online]. Available: <https://doi.org/10.2196/57760>
- [107] Toor et al., “Optimizing the surgical instrument tray to immediately increase efficiency and lower costs in the operating room,” *Canadian Journal of Surgery*, 2022. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9007441/>
- [108] D.L. Kirkpatrick, *Evaluating Training Programs: The Four Levels*. San Francisco: Berrett-Koehler, 1994, the four-level training evaluation model: Reaction, Learning, Behavior, Results. Level 2 (Learning) measures knowledge/skill acquisition; Level 3 (Behavior) measures on-the-job transfer.
- [109] S.M. Downing, “Validity: on the meaningful interpretation of assessment data,” *Medical Education*, vol. 37, no. 9, pp. 830–837, 2003, validity as a unitary concept supported by five sources of evidence in medical education assessment.
- [110] M.T. Kane, “Validating the interpretations and uses of test scores,” *Journal of Educational Measurement*, vol. 50, no. 1, pp. 1–73, 2013, argument-based validity framework: the validity of a test-score interpretation depends on the plausibility of the interpretive argument, including assumptions about the target domain, scoring, generalization, extrapolation, and implications.
- [111] Justin Kruger and David Dunning, “Unskilled and unaware of it: How difficulties in recognizing one’s own incompetence lead to inflated self-assessments,” *Journal of Personality and Social Psychology*, vol. 77, no. 6, pp. 1121–1134, 1999.
- [112] Chobin, “The real costs of surgical instrument training in sterile processing revisited,” *AORN Journal*, 2010. [Online]. Available: <https://doi.org/10.1016/j.aorn.2009.10.025>
- [113] Natali et al., “Ai-induced deskilling in medicine: A mixed-method review and research agenda for healthcare and beyond,” *Artificial Intelligence Review*, 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s10462-025-11352-1>
- [114] Zheng et al., “Designing human-centered ai to prevent medication dispensing errors: Focus group study with pharmacists,” *JMIR Formative Research*, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10775023/>
- [115] Amershi et al., “Guidelines for human-ai interaction,” *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019. [Online]. Available: <https://www.microsoft.com/en-us/research/publication/guidelines-for-human-ai-interaction/>

- [116] Olakotan and Yusof, “The appropriateness of clinical decision support systems alerts in supporting clinical workflows: A systematic review,” *Health Informatics Journal*, 2021. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/33853395/>
- [117] Cánovas-Segura et al., “Meaningful time-related aspects of alerts in clinical decision support systems. a unified framework,” *Journal of Biomedical Informatics*, 2023. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/37245656/>
- [118] Xin et al., “Few-shot object detection: Research advances and challenges,” *Information Fusion*, 2024. [Online]. Available: <https://doi.org/10.1016/j.inffus.2024.102307>
- [119] Rodrigues et al., “Evaluation of deep learning techniques on a novel hierarchical surgical tool dataset,” *AI 2021: Advances in Artificial Intelligence*, 2022. [Online]. Available: <https://hdl.handle.net/10289/14991>
- [120] Fayad et al., “Traceability of surgical instruments: A systematic review,” *Applied Sciences*, 2025. [Online]. Available: <https://doi.org/10.3390/app15031592>
- [121] Vithlani et al., “Economic evaluations of artificial intelligence-based healthcare interventions: a systematic literature review of best practices in their conduct and reporting,” *Frontiers in Pharmacology*, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10486896/>
- [122] Kastrup et al., “Landscape and challenges in economic evaluations of artificial intelligence in healthcare: a systematic review of methodology,” *BMC Digital Health*, 2024. [Online]. Available: <https://bmcdigitalhealth.biomedcentral.com/articles/10.1186/s44247-024-00088-7>

A Research Support Audit

Table 38: Research support audit: design choice traceability

Design Choice	Research Support	Defensible Stance
Time-to-competency as the north star	SPD certification and training evidence shows that entry into practice requires hands-on hours, structured training, and clinical supervisor support [22]; [112]; [25].	Measure simulated improvement in local tray familiarity and report the longitudinal onboarding question as future field validation.
Retrieval-centered learning loop	Simulation, retrieval practice, spacing, feedback, and pre/post testing are supported by health-professions and learning-science evidence, with direct SP training precedent in Ofstead et al. and digital-practice support from spaced education and electronic-flashcard research [24]; [66]; [86]; [94]; [106]; [69]; [96].	Build pre-test, retrieval-first cards, quizzes, applied tray practice with feedback, weak-item review, post-test, and metrics export as one module.
Local tray authoring	Instrument families are reusable, but actual tray requirements are local: count sheets specify tray contents, quantities, sizes, and catalog/reference numbers; tray optimization depends on procedure, surgeon preference, usage likelihood, and stock decisions; specialty and loaner trays can be vendor-specific [23]; [16]; [17].	Start with file-backed instructor-verified modules and record local names, aliases, quantities, photos, and version history.
Human authority	Healthcare AI literature warns about automation bias, deskilling, liability, and workflow fit [87]; [113]; [88]; [114].	The system may teach, score, explain, and log; instructors and supervisors remain responsible for real competency judgments.
Workflow-first learner UI	General usability guidance supports visible status, efficient action paths, error prevention, and recoverable mistakes. Human-AI guidance supports efficient invocation, dismissal, and correction when AI guesses wrong. Clinical alert literature warns that low-value alerts and added tasks can undermine acceptance [115]; [116]; [117].	Design around the learning mode, use clear mode boundaries, provide fast practice feedback, and suppress hints during assessment.
CV feasibility as an external assumption	Few-shot object detection and surgical-instrument CV papers show that data-efficient detection and constrained instrument recognition are plausible [118]; [80]; [61].	Cite existing papers for feasibility and reserve local CV accuracy for future support-layer validation.
CV generalization as a future validation problem	Kienle et al. show a large cross-manufacturer performance drop despite strong in-domain instrument-stand results [81].	Any future camera-supported feature needs site-specific validation. The current project can function through manual simulated sorting.
Lookalike tools as learning content	HOSPITools and Atabuzzaman et al. highlight subtle instrument	Teach distinguishing features in retrieval-first cards and track

B Implementation Defaults

Table 39: Implementation defaults and critique-ready interpretation

Default	Current Use	Critique-Ready Interpretation
File-backed tray and instrument data	Fast local authoring for one seeded module.	Good for a capstone project. A real training program would need role permissions, content review, versioning, and integration decisions.
Pre/post simulated tray sorting	Primary training-effectiveness evidence.	Measures simulated local familiarity and produces paired learning metrics.
Confidence ratings	Learner metacognition and instructor review.	Useful for spotting overconfidence and fragile knowledge.
Existing YOLO/Streamlit tray-check code	Demo code and possible future visual support.	Useful as support-layer infrastructure after separate validation.
Local ignored <code>data/</code> , <code>runs/</code> , and <code>weights/</code> folders	Keeps raw images, training outputs, and weights out of git.	Sensible prototype hygiene. Formal deployment would require access controls, retention policy, audit policy, and privacy review.

C CV Experiment Methods and Additional Results

C.1 Brightness Experiment Setup

The brightness experiments used a self-collected dataset of four spoon variants and one fork (five classes) photographed on two backgrounds (matte white, reflective silver) across 11 brightness levels forming a descending ladder from 800 to 80 estimated lumens. Two layout conditions were captured: separated (instruments arranged in two distinct orders) and overlay (instruments overlapping in a single arrangement). Seven staged train/test splits were constructed from these conditions to isolate the effect of each variable independently. Each stage was trained with YOLO11s for 100 epochs at 640×640 resolution.

Figure 3 shows example images from the separated training set and the overlay test set, illustrating the layout gap that produced the largest performance degradation.



(a) Separated layout (training condition)



(b) Overlay layout (test condition)

Figure 3: Example images from the separated-to-overlay experiment. The training set uses non-overlapping layouts; the test set uses overlapping layouts.

C.2 Real-Instrument Experiments

The surgical kit proxy dataset contains six real instrument classes (scissor1, scissor2, scissor3, scissor4, forcep, scalpel) photographed on matte and reflective backgrounds across four sessions. Each image contains exactly six instruments (one per class) in fixed positions per session. Three staged experiments were trained with YOLO11s for 100 epochs.

Table 40 shows the per-class AP50 breakdown. The pattern is diagnostic of position-cue learning: classes that achieve high AP in one split fall to zero in others, matching session-specific instrument positions rather than visual features.

Table 40: Per-class AP50 for the three real-instrument experiments

Class	matte→reflective	reflective→matte	shape_similarity
scissor1	0.000	0.000	0.862
scissor2	0.435	0.000	0.000
scissor3	0.000	0.000	0.000
scissor4	0.000	0.000	0.587
forcep	0.517	0.000	0.000
scalpel	0.446	0.717	0.069

Figure 4 shows the most striking failure mode: the `reflective_train_matte_test` model detects only the scalpel (the most visually distinct instrument) and misses all five remaining classes. The confusion matrix for the `shape_similarity_real_proxy` experiment (Figure 5) confirms that class associations are split-specific rather than shape-driven.

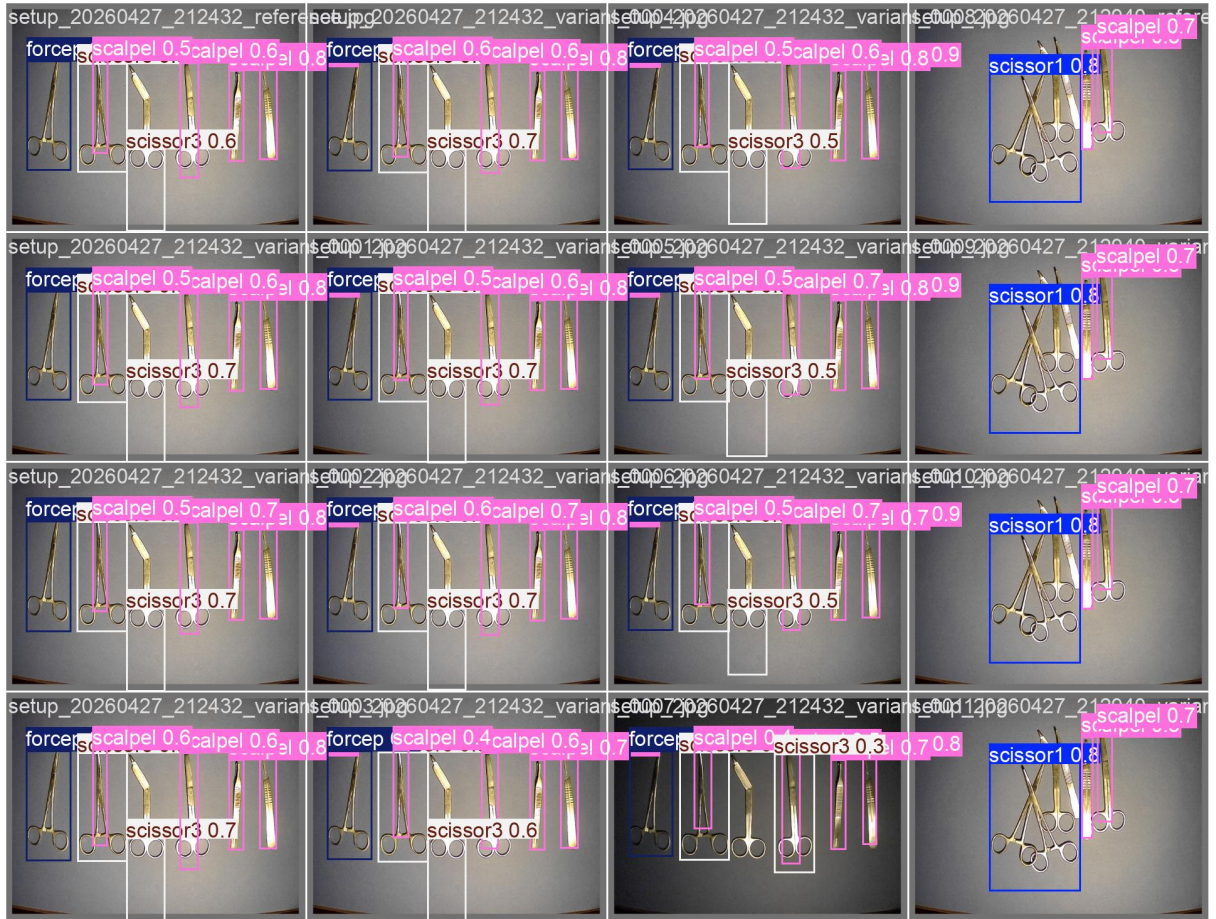


Figure 4: Prediction example from `reflective_train_matte_test`. Only the scalpel is detected (bounding box); the other five instruments (scissor1–4, forcep) are all missed.

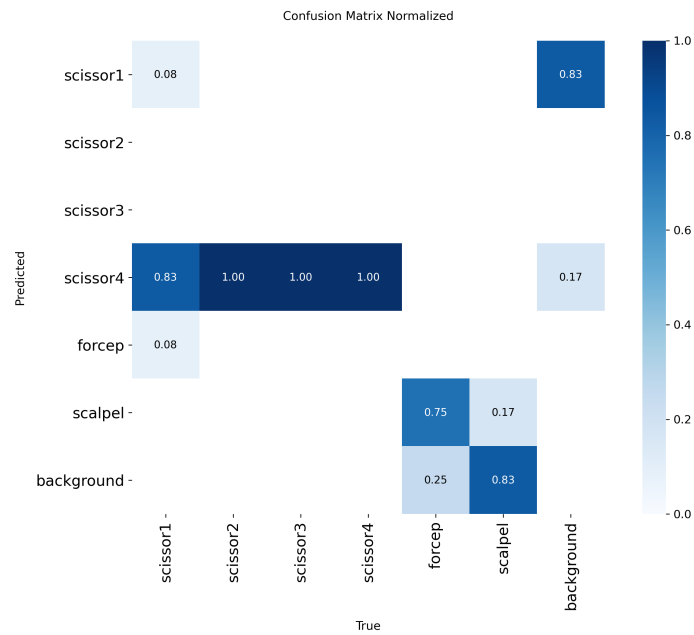


Figure 5: Normalized confusion matrix for `shape_similarity_real_proxy`. `scissor1` and `scissor4` are the only classes with nonzero true-positive rates; the remaining classes are uniformly missed.

Appendix D: Bills of Materials

Table 41: Bills of materials

Item	Description	Qty	Unit	Supplier	Unit Cost	Extended
1	Cardstock paper (letter, 65 lb)	1	pack (50)	Office supply	\$8	\$8
2	Clear adhesive sleeves	1	pack (100)	Office supply	\$10	\$10
3	Foam backing board (20×30 in)	1	sheet	Craft store	\$5	\$5
4	AprilTag markers (printed on paper)	30	sheets	In-house	\$0	\$0
5	Laptop or tablet for scanning	1	unit	Existing	\$0	\$0
6	Camera stand (optional)	1	unit	Online	\$20	\$20
7	Printed study flashcards	30	sheets	In-house	\$0	\$0
8	Assorted office supplies (tape, scissors, binder clips)	1	set	Office supply	\$10	\$10
Total prototype material cost						\$53

The primary project cost is labor: module authoring, software implementation, experimental data collection, study moderation, analysis, and report writing. These are captured in the project plan’s task assignments and are not itemized as direct material purchases. The prototype is designed to work with materials already available in a typical academic or office environment. Costs above are upper-bound estimates; actual spending can be lower by reusing existing equipment and printing on available paper.

Appendix E: Collaboration Task Matrix

Table 42: Collaboration task matrix

Team Member	Sections Authored	Key Contributions
Matthew Li	Executive Summary, Introduction, Problem Definition and Scope (with Andy, Qiyu), Preliminary Analysis, First Attempt (with Qiyu), Literature and Related Work (with Andy, Qiyu), Final Product Concept (with Andy, Qiyu), Requirements Definition (with Andy, Jacob), System Design (with Jacob), Analysis (with Qiyu), Design Outcome vs Requirements (with Qiyu), Evaluation and Validation Plan (with Jacob), Risk Assessment (with Jacob), Broader Impacts (with Jacob), Project Plan (with Jacob), Discussion (with Jacob), Research Support Audit, Implementation Defaults	Root-cause analysis, preliminary analysis (all 6 methods), requirements traceability matrix, training web app implementation, tray module data model, scoring engine, card generation pipeline, export and run storage, CV data collection and YOLO training pipeline, multi-variant benchmark harness, unit tests, study design
Andy Ma	Problem Definition and Scope (with Matthew, Qiyu), Literature and Related Work (with Matthew, Qiyu), Final Product Concept and Design Rationale (with Matthew, Qiyu), Requirements Definition: Element Definition and Objective Tree (with Matthew)	Literature review, problem explication, related work research, product concept, objective tree development
Qiyu Xin	Problem Definition and Scope (with Matthew, Andy), First Attempt (with Matthew), Literature and Related Work (with Matthew, Andy), Final Product Concept (with Matthew, Andy), Analysis (with Matthew), Design Outcome vs Requirements (with Matthew)	Computer-vision experiments (brightness, occlusion, real-instrument studies), literature review, product concept and design rationale
Jacob Moore	Requirements Definition: Performance Specifications, Quality Function Deployment, and Requirement Traceability Matrix (with Matthew, Andy), System Design (with Matthew), Evaluation and Validation Plan (with Matthew), Risk Assessment (with Matthew), Broader Impacts (with Matthew), Project Plan (with Matthew), Discussion (with Matthew), References	Performance specifications, QFD, RTM, system design, evaluation framework, risk register, ethical analysis, project plan postmortem, budget, references list curation
Owen Fan	(Appendices support)	Experimental data collection assistance, figure generation, documentation review